Carl Friedrich Gauss

Works, Volume IV

Theory of the Combination of Observations Subject to the Smallest Errors

First Part, Second Part and Supplement

AI-translated into English from the historical Latin and German. The German volume titles are AI-translated into English. This reader contains the historical volume titles, the formal First Part title and the complete First Part, Articles 1–22, including both author footnotes, and the formal and repeated titles of the complete Second Part, Articles 23–40, including the author footnote. The complete Supplement follows, with its formal and repeated titles, Articles 1–25 and all four author footnotes. The complete notice of 26 February 1821 from the Göttingische gelehrte Anzeigen is also included. The other notices and works in Volume IV are outside this reader.

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Translation sources and terminology

The Latin text is from Carl Friedrich Gauss, *Werke*, Band IV (Göttingen, 1873), issued by the Royal Society of Sciences at Göttingen. The original-language editable transcription and its historical portrait are available in the original-language edition. The modern source encoding and typesetting are credited to OpenAI GPT-6 Astra, Ultra; this English translation and its typesetting are AI-produced.

Related English mathematical-argument usage was consulted in James Caddall Morehead and Adam Miller Hiltebeitel's *General Investigations of Curved Surfaces of 1827 and 1825* (1902), Project Gutenberg 36856, TeX edition. Related German quantity and mathematical-argument usage was consulted in H. Maser's translation and edition, *Carl Friedrich Gauss' Untersuchungen über höhere Arithmetik* (1889), historical digitized edition. These works are not parallel translations of the observation theory. The German notice of 26 February 1821 (pp. 100–105) is translated here in full; it also supplied related error and probability terminology for the First Part. No human translation prose or formulas are taken from Morehead–Hiltebeitel or Maser. Historical publication titles remain in the original. Exact versions and URLs are retained in `CITATIONS.bib`.

For the Second Part, related English expression, substitution and mathematical-argument usage in Morehead and Hiltebeitel and related German quantity, substitution and equation usage in Maser were consulted. These are not parallel translations, attestations of the historical wording or authority for the formulas or proof. No human running prose or modern formulas are adopted. The German notice of 1821 was consulted for the First Part only. The Second Part was presented on 2 February 1823 and published in 1823.

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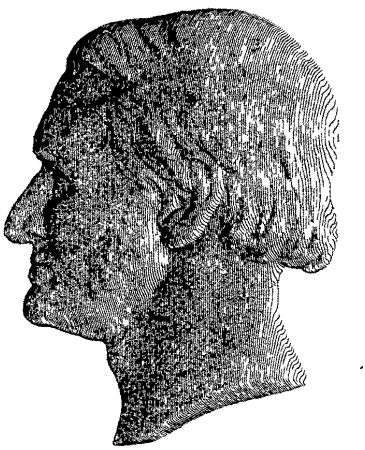
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CARL FRIEDRICH GAUSS

WORKS

FOURTH VOLUME.



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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS

FIRST PART

BY

CARL FRIEDRICH GAUSS

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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS.

First Part.

1.

However much care is taken in making observations concerned with the magnitudes of natural things, they nevertheless always remain subject to larger or smaller errors. Errors of observation are usually not simple, but arise from several sources at once: two kinds of these sources must be carefully distinguished. Some causes of error are such that their effect in any observation depends on variable circumstances between which and the observation itself no essential connection is understood: the errors arising from these causes are called irregular or random, and insofar as those circumstances cannot be subjected to calculation, the same holds for the errors themselves. Such are the errors arising from the imperfection of the senses, and also from irregular external causes, e.g. the quivering motion of the air, which slightly disturbs vision: several defects of even the best instruments must also be included here, e.g. roughness of the inside of spirit levels, lack of absolute stability, etc. Other causes of error, by contrast, by their nature produce in all observations of the same kind an effect that is either absolutely constant, or at least one whose magnitude depends, according to a definite law, solely on circumstances regarded as essentially connected with the observation. Errors of this kind are called constant or regular.

It is evident, moreover, that this distinction is relative to some extent, and depends on the wider or narrower sense in which the notion of observations belonging to the same kind is taken. For example, irregular defects in the graduation of instruments for measuring angles produce a constant error whenever the question is merely one of repeating the observation of a particular angle indefinitely, since the same faulty divisions are always used in this case: on the other hand, the error arising from this source can be regarded as random whenever the question is indefinitely one of measuring angles of any magnitude, provided there is no table showing the amount of error in the individual divisions.

2.

Strictly speaking, the consideration of regular errors is excluded from our purpose. It is the observer's task to investigate diligently all causes capable of producing constant errors, and either to remove them, or at least to examine

their nature and magnitude with the utmost care, so that their effect can be determined in any particular observation. The observation can thus be freed from that effect, which amounts to the same thing as if the error had not been present at all. The situation is very different, however, with irregular errors, which by their nature cannot be subjected to calculation. These must therefore be tolerated in the observations, but their effect on the quantities to be derived from the observations must be reduced as much as possible by a skilful combination of the latter. The following investigations are devoted to this most important subject.

3.

Errors in observations belonging to the same kind that arise from a particular simple cause are, by the nature of the matter, confined within certain *limits*, which could undoubtedly be assigned exactly if the nature of the cause itself were *fully* understood. Most causes of random errors are such that, according to the law of continuity, all errors contained within those limits must be regarded as possible, and perfect knowledge of the cause would also teach us whether all these errors are equally likely or not equally likely. In the latter case it would also teach us what relative probability is to be assigned to each error. The same will hold for the total error, compounded from several simple errors: namely, it will be enclosed within certain limits, (one of which will be equal to the sum of all the partial upper limits, the other to the sum of all the lower limits); all errors within these limits will indeed be possible, but since each can be composed in infinitely many different ways from partial errors that are themselves more or less probable, we shall have to assign a greater chance to one and a smaller chance to another, and the law of relative probability can be found if the laws of the simple errors are assumed known, subject to the analytical difficulties of collecting all the combinations.

There are, indeed, some causes of error that can produce only discrete errors, rather than errors proceeding according to the law of continuity, such as errors in the graduation of instruments, (if one chooses to count these among random errors): for the number of divisions in any particular instrument is finite. It is evident, however, that notwithstanding this, provided that not all the causes of error produce discrete errors, the collection of all possible total errors will constitute a series proceeding according to the law of continuity, or several such interrupted series if, when all possible discrete errors are arranged in order of magnitude, one or another difference between two adjacent terms happens to be greater than the difference between the limits of the total errors insofar as they arise from continuous errors alone. But in practice the latter case will hardly ever occur, unless the graduation suffers from rather gross defects.

4.

If we denote the relative likelihood of the total error x, in a particular kind of observation, by the characteristic φx, then, because of the continuity of the errors, this is to be understood as meaning that the probability of an error between the infinitely close limits x and $x + \mathrm{d}x$ is $= \varphi x.\mathrm{d}x$. In practice it will hardly ever, indeed scarcely ever at all, be possible to assign this function a priori: nevertheless several general results concerning it can be established, which we shall present below. The following is evident. The function φx must be classed among discontinuous functions insofar as, for all values of x lying outside the limits of the possible errors, it must be $= 0$; within these limits, however, it will take a positive value everywhere (leaving aside the case of which we spoke at the end of the preceding article). In most cases it will be permissible to suppose positive and negative errors of the same magnitude equally likely, in which case $\varphi(-x) = \varphi x$. Further, since smaller errors are more readily committed than larger ones, the value of φx will usually be greatest for $x = 0$, and will decrease continuously as x increases.

In general, however, the value of the integral $\int \varphi x.\mathrm{d}x$, extended from $x = a$ to $x = b$, will express the probability that an error not yet known lies between the limits a and b. The value of this integral from the lower limit of all possible errors to the upper limit will therefore always be $= 1$. And since φx is, for all values of x outside these limits, always $= 0$, it is evident also that

the value of the integral $\int \varphi x.\mathrm{d}x$ *extended from* $x = -\infty$ *to* $x = +\infty$ *is always* $= 1$.

5.

Let us further consider the integral $\int x\varphi x.\mathrm{d}x$ between the same limits, whose value we shall put $= k$. If all the simple causes of error are such that there is no reason why one of two errors equal in magnitude but opposite in sign should be produced more readily than the other, this will also hold for the total error, or we shall have $\varphi(-x) = \varphi x$, and consequently, necessarily, $k = 0$. From this we conclude that whenever k does not vanish, but is, for example, a positive quantity, there must necessarily be one or another cause of error that can either produce positive errors only, or at least produce positive errors more readily than negative ones. This quantity k, which is in fact the mean of all possible errors, or the mean value of x, can conveniently be called the constant part of the error. Moreover, it can easily be proved that the constant part of the total error is equal to the sum of the constant parts contained in the errors arising from the individual simple causes. If the quantity k is supposed known and is subtracted from every observation, and if the error of the observation thus corrected is denoted by x', and its probability by $\varphi' x'$, we shall have $x' = x - k$, $\varphi' x' = \varphi x$ and consequently $\int x'\varphi' x'.\mathrm{d}x' = \int x\varphi x.\mathrm{d}x - \int k\varphi x.\mathrm{d}x = k - k = 0$, i.e. the errors of the

corrected observations will have no constant part, which is also evident in itself.

6.

Just as the integral $\int x\varphi x.\mathrm{d}x$, or the mean value of x, tells us either the absence of a constant error or its presence and magnitude, the integral

$$\int xx\varphi x.\mathrm{d}x$$

extended from $x = -\infty$ to $x = +\infty$ (or the mean value of the square xx) seems best suited to defining and measuring the uncertainty of observations in general, so that, of two systems of observations differing in the likelihood of their errors, those in which the integral $\int xx\varphi x.\mathrm{d}x$ has the smaller value are to be judged superior in precision. If anyone objects that this way has been chosen arbitrarily, without any compelling necessity, we shall readily agree. For this question, by its nature, involves something indefinite that cannot be confined within bounds except by a principle that is to some extent arbitrary. The determination of a quantity by an observation subject to a larger or smaller error may aptly be compared to a game in which there are losses only and no gains, since every error to be feared is akin to a loss. The cost of such a game is assessed by the probable loss, namely by the sum of the products of the individual possible losses and their respective probabilities. But what loss should be taken as equivalent to each error of observation is by no means evident in itself; rather, this determination depends to some extent on our choice. It is evidently not permissible to put the loss equal to the error itself; for if positive errors were taken as losses, negative ones would have to represent gains. The magnitude of the loss must instead be expressed by a function of the error that by its nature is always positive. Since the variety of such functions is infinite, the simplest that has this property seems to be preferred to the others, and this is indisputably the square: in this way the principle stated above emerges.

The illustrious Laplace did indeed consider the matter in a similar way, but adopted the error itself, always taken positively, as the measure of loss. But unless we are mistaken, this way is at least no less arbitrary than ours: for whether a double error is considered as tolerable as a single one repeated twice, or less tolerable, and consequently whether it is more appropriate to assign to a double error a weight only twice as great or a greater one, is a question neither evident in itself nor to be decided by mathematical demonstrations, but one to be left solely to free choice. Furthermore, it cannot be denied that continuity is violated in that way: and for this very reason that method is less amenable to analytical treatment, whereas the results to which our principle leads commend themselves by their remarkable simplicity as well as generality.

7.

Putting the value of the integral $\int xx\varphi x.\mathrm{d}x$ extended from $x = -\infty$ to $x = +\infty = mm$, we shall call the quantity m the *mean error to be feared*, or simply the *mean error*, of observations whose indeterminate errors x have the relative probability φx. We shall not restrict this name to direct observations, but shall also extend it to any determinations derived from observations. Care must be taken, however, not to confuse the mean error with the arithmetic mean of all the errors, of which we spoke in art. 5.

Suppose several kinds of observations, or several determinations obtained from observations, to which the same precision cannot be assigned, are being compared. Their relative *weight* for us will be a quantity inversely proportional to mm, while their *precision* is taken as simply inversely proportional to m. In order, therefore, that a weight may be expressed by a number, the weight of a particular kind of observation must have been taken as the unit.

8.

If the errors of observations contain a constant part, removing it diminishes the mean error and increases the weight and precision. Retaining the notation of art. 5, and denoting the mean error of the corrected observations by m', we shall have

$$\begin{aligned} m'm' &= \int x'x'\varphi'x'.\mathrm{d}x' = \int (x-k)^2\varphi x.\mathrm{d}x \\ &= \int xx\varphi x.\mathrm{d}x - 2k\int x\varphi x.\mathrm{d}x + kk\int \varphi x.\mathrm{d}x \\ &= mm - 2kk + kk = mm - kk. \end{aligned}$$

If, however, instead of the true constant part k, another quantity l had been subtracted from the observations, the square of the new mean error would become $= mm - 2kl + ll = m'm' + (l-k)^2$.

9.

If λ denotes a definite coefficient, and μ the value of the integral $\int \varphi x.\mathrm{d}x$ from $x = -\lambda m$ to $x = +\lambda m$, then μ will be the probability that the error of an observation is less than λm (without regard to sign), and $1-\mu$ the probability of an error greater than λm. If, therefore, the value $\mu = \frac{1}{2}$ corresponds to the value $\lambda m = \rho$, the error can fall below ρ just as readily as above ρ, so that ρ can conveniently be called the *probable* error. The relation between the quantities λ, μ evidently depends on the nature of the function φx, which is usually unknown. It will therefore be worthwhile to consider this relation more closely in certain special cases.

I. If the limits of all possible errors are $-a$ and $+a$, and all errors within these limits are equally probable, φx will be constant between the limits $x = -a$ and $x = +a$, and consequently $= \frac{1}{2a}$. Hence $m = a\sqrt{\frac{1}{3}}$, and also

$\mu = \lambda\sqrt{\frac{1}{3}}$, as long as λ is not greater than $\sqrt{3}$; finally, $\rho = m\sqrt{\frac{3}{4}} = 0{,}8660254m$, and the probability that an error no greater than the mean error occurs will be $= \sqrt{\frac{1}{3}} = 0{,}5773503$.

II. If, as before, $-a$ and $+a$ are the limits of the possible errors, and the probability of the errors themselves is supposed to decrease on both sides from the error 0 in arithmetic progression, we shall have

$$\varphi x = \frac{a - x}{aa}, \quad \text{for values of } x \text{ between 0 and } +a$$
$$\varphi x = \frac{a + x}{aa}, \quad \text{for values of } x \text{ between 0 and } -a$$

From this we deduce $m = a\sqrt{\frac{1}{6}}$, $\mu = \lambda\sqrt{\frac{2}{3}} - \frac{1}{6}\lambda\lambda$, as long as λ is between 0 and $\sqrt{6}$; finally, $\lambda = \sqrt{6} - \sqrt{(6 - 6\mu)}$, as long as μ is between 0 and 1, and consequently

$$\rho = m(\sqrt{6} - \sqrt{3}) = 0{,}7174389m$$

The probability of an error not exceeding the mean error will in this case be

$$= \sqrt{\frac{3}{2}} - \frac{1}{6} = 0{,}6498299$$

III. If we put the function φx proportional to $e^{-\frac{xx}{hh}}$ (which can indeed be only approximately true in nature), we must have

$$\varphi x = \frac{e^{-\frac{xx}{hh}}}{h\sqrt{\pi}}$$

where π denotes the semicircumference of a circle of radius 1, whence we further deduce

$$m = h\sqrt{\frac{1}{2}}$$

(See *Disquis. generales circa seriem infinitam* etc. art. 28). Further, if the value of the integral

$$\frac{2}{\sqrt{\pi}} \int e^{-zz} \mathrm{d}z$$

beginning at $z = 0$ is denoted by Θz, we shall have

$$\mu = \Theta(\lambda\sqrt{\tfrac{1}{2}})$$

The following table gives some values of this quantity:

λ	μ
0,6744897	0,5
0,8416213	0,6
1,0000000	0,68268·95
1,0364334	0,7
1,2815517	0,8
1,6448537	0,9
2,5758293	0,99
3,2918301	0,999
3,8905940	0,9999
∞	1

10.

Although the relation between λ and μ depends on the nature of the function φx, some general results can nevertheless be established. Namely, whatever this function may be, provided it is such that its value always decreases, or at least does not increase, as the absolute value of x increases, we shall certainly have

> λ less than, or at least not greater than, $\mu\sqrt{3}$, whenever μ is less than $\frac{2}{3}$;
>
> λ not greater than $\frac{2}{3\sqrt{(1-\mu)}}$, whenever μ is greater than $\frac{2}{3}$.

For $\mu = \frac{2}{3}$ the two limits coincide: namely, λ cannot be greater than $\sqrt{\frac{4}{3}}$.

To prove this remarkable theorem, let us denote by y the value of the integral $\int \varphi z.\mathrm{d}z$ extended from $z = -x$ to $z = +x$, so that y will be the probability that an error is contained within the limits $-x$ and $+x$. Further, let us put

$$x = \psi y, \quad \mathrm{d}\psi y = \psi' y.\mathrm{d}y, \quad \mathrm{d}\psi' y = \psi'' y.\mathrm{d}y$$

We shall therefore have $\psi 0 = 0$, and also

$$\psi' y = \frac{1}{\varphi x + \varphi(-x)}$$

so that, by hypothesis, $\psi' y$ will always increase from $y = 0$ to $y = 1$, or at least will nowhere decrease; or, what is the same, the value of $\psi'' y$ will always be positive, or at least not negative. Further, we have $\mathrm{d}.y\psi' y = \psi' y.\mathrm{d}y + y\psi'' y.\mathrm{d}y$, and consequently

$$y\psi' y - \psi y = \int y\psi'' y.\mathrm{d}y$$

the integration beginning at $y = 0$. The value of the expression $y\psi' y - \psi y$ will therefore always be a positive quantity, or at least not a negative one, and consequently

$$1 - \frac{\psi y}{y\psi' y}$$

will be a positive quantity less than unity. Let f be its value for $y = \mu$, i.e. since we have $\psi\mu = \lambda m$, let

$$f = 1 - \frac{\lambda m}{\mu\psi'\mu} \quad \text{or} \quad \psi'\mu = \frac{\lambda m}{(1-f)\mu}$$

With these preparations, let us consider the following function of y

$$\frac{\lambda m}{(1-f)\mu}(y - \mu f)$$

which we shall put $= Fy$, and also $\mathrm{d}Fy = F'y.\mathrm{d}y$. It is evident that

$$F\mu = \lambda m = \psi\mu$$

$$F'\mu = \frac{\lambda m}{(1-f)\mu} = \psi'\mu$$

Since, therefore, $\psi'y$ increases continuously as y increases (or at least does not decrease, which is always to be understood), while $F'y$ is constant, the difference $\psi'y - F'y = \frac{\mathrm{d}(\psi y - Fy)}{\mathrm{d}y}$ will be positive for values of y greater than μ, and negative for smaller ones. From this it is easily concluded that $\psi y - Fy$ is always a positive quantity, and consequently ψy will always be absolutely greater, or at least not smaller, than Fy, certainly as long as the value of Fy is positive, i.e. from $y = \mu f$ to $y = 1$. Hence the value of the integral $\int (Fy)^2 \mathrm{d}y$ from $y = \mu f$ to $y = 1$ will be smaller than the value of the integral $\int (\psi y)^2 \mathrm{d}y$ between the same limits, and therefore still more certainly smaller than the value of this integral from $y = 0$ to $y = 1$, which is $= mm$. But the value of the former integral is found to be

$$= \frac{\lambda\lambda mm(1-\mu f)^3}{3\mu\mu(1-f)^2}$$

whence we conclude that $\lambda\lambda$ is less than $\frac{3\mu\mu(1-f)^2}{(1-\mu f)^3}$, where f is a quantity lying between 0 and 1. Now the value of the fraction $\frac{3\mu\mu(1-f)^2}{(1-\mu f)^3}$, whose differential, if f is regarded as a variable quantity, is $=$

$$-\frac{3\mu\mu(1-f)}{(1-\mu f)^4}.(2-3\mu+\mu f)\mathrm{d}f$$

decreases continuously as f passes from the value 0 to the value 1, whenever μ is less than $\frac{2}{3}$; consequently the greatest possible value will be that corresponding to $f = 0$, namely $= 3\mu\mu$, so that in this case λ is certainly less than or not greater than $\mu\sqrt{3}$. Q. E. P. On the other hand, whenever μ is greater than $\frac{2}{3}$, the value of that fraction will be greatest for $2 - 3\mu + \mu f = 0$, i.e. for $f = 3 - \frac{2}{\mu}$, whence it becomes $= \frac{4}{9(1-\mu)}$, and therefore in this case λ is not greater than $\frac{2}{3\sqrt{(1-\mu)}}$. Q E. S.

Thus, for example, for $\mu = \frac{1}{2}$, λ certainly cannot be greater than $\sqrt{\frac{3}{4}}$, i.e. the probable error cannot exceed the limit $0{,}8660254m$, to which it was found equal in the first example of art. 9. Further, it is easily concluded from our theorem that μ is not less than $\lambda\sqrt{\frac{1}{3}}$ as long as λ is less than $\sqrt{\frac{4}{3}}$; on the other hand, μ is not less than $1 - \frac{4}{9\lambda\lambda}$ for a value of λ greater than $\sqrt{\frac{4}{3}}$.

11.

Since several problems to be treated below are also connected with the value of the integral $\int x^4 \varphi x . \mathrm{d}x$, it will be worthwhile to develop it for certain special cases. We shall denote the value of this integral extended from $x = -\infty$ to $x = +\infty$ by n^4.

I. For $\varphi x = \frac{1}{2a}$, insofar as x is contained between $-a$ and $+a$, we have $n^4 = \frac{1}{5}a^4 = \frac{9}{5}m^4$.

II. In the second case of art. 6, where $\varphi x = \frac{a \mp x}{aa}$, for values of x between 0 and $\pm a$, we obtain $n^4 = \frac{1}{15}a^4 = \frac{12}{5}m^4$.

III. In the third case, where

$$\varphi x = \frac{e^{-\frac{xx}{hh}}}{a\sqrt{\pi}}$$

we find, by what is set out in the memoir cited above, $n^4 = \frac{3}{4}h^4 = 3m^4$.

Moreover, it can be proved that the value of $\frac{n^4}{m^4}$ is certainly not less than $\frac{9}{5}$, provided that the hypothesis of the preceding article holds.

12.

Let x, x', x'' etc. denote indefinitely the mutually independent errors of observations of the same kind, whose relative probabilities are expressed by the prefixed characteristic φ; and let y denote a given rational function of the indeterminates x, x', x'' etc.: the multiple integral (I)

$$\int \varphi x.\varphi x'.\varphi x'' \ldots\ldots \mathrm{d}x.\mathrm{d}x'.\mathrm{d}x'' \ldots\ldots$$

extended over all values of the indeterminates x, x', x'' for which the value of y falls within the given limits 0 and η will express the probability of a value of y lying indefinitely within 0 and η. Evidently this integral will be a function of η, whose differential we shall put $= \psi\eta.\mathrm{d}\eta$, so that the integral itself becomes equal to the integral $\int \psi\eta.\mathrm{d}\eta$ beginning at $\eta = 0$. In this way the characteristic $\psi\eta$ must at the same time be regarded as expressing the relative probability of any value of y. Since x can be regarded as a function of the indeterminates y, x', x'' etc., which we shall put

$$= f(y, x', x'' \ldots.)$$

the integral (I) will become

$$= \int \varphi.f(y, x', x'' \ldots).\frac{\mathrm{d}f(y, x', x'' \ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x'' \ldots. \mathrm{d}y.\mathrm{d}x'.\mathrm{d}x'' \ldots.$$

where y must be extended from $y = 0$ to $y = \eta$, while the remaining indeterminates must range over all values to which a real value of $f(y, x', x'' \ldots.)$ corresponds. From this it follows that

$$\psi y = \int \varphi.f(y, x', x'' \ldots.).\frac{\mathrm{d}f(y, x', x'' \ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x'' \ldots. \mathrm{d}x'.\mathrm{d}x'' \ldots.$$

the integration, in which y must be regarded as constant, extending over all values of the indeterminates x', x'' etc. that give $f(y, x', x'' \ldots.)$ a real value.

13.

To carry out this integration in practice, knowledge of the function φ, which is usually unknown, would be required: indeed, even if this function were known, in most cases the integration would exceed the powers of analysis. This being so, we shall be unable to assign the probability of the individual values of y: but the case will be different if only the mean value of y is wanted, which arises from the integration of $\int y\psi y.\mathrm{d}y$ extended over all values of y

that it can attain. And since, evidently, for all values that y cannot attain, either because of the nature of the function it expresses (e.g. for negative values, if we had $y = xx + x'x' + x''x'' + \text{etc.}$), or because certain limits are set to the errors x, x', x'' etc. themselves, we must put $\psi y = 0$, the result will evidently be the same if that integration is extended over all real values of y, namely from $y = -\infty$ to $y = +\infty$. Now the integral $\int y\psi y.\mathrm{d}y$ taken between definite limits, namely from $y = \eta$ to $y = \eta'$, is equal to the integral

$$\int y\varphi.f(y, x', x''\ldots.)\frac{\mathrm{d}f(y, x', x''\ldots.)}{\mathrm{d}y}. \varphi x'.\varphi x''\ldots.\mathrm{d}y.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

the integration extending from $y = \eta$ to $y = \eta'$ and over all values of the indeterminates x', x'' etc. to which a real value of $f(y, x', x''\ldots.)$ corresponds; or, what is the same, to the value of the integral

$$\int y\varphi x.\varphi x'.\varphi x''\ldots.\mathrm{d}x.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

using in this integration for y its value expressed in terms of x, x', x'' etc., and extending the integration over all values of these indeterminates to which a value of y lying between η and η' corresponds. From this we conclude that the integral $\int y\psi y.\mathrm{d}y$ extended over all values of y, from $y = -\infty$ to $y = +\infty$, is obtained from the integration of

$$\int y\varphi x.\varphi x'.\varphi x''\ldots.\mathrm{d}x.\mathrm{d}x'.\mathrm{d}x''\ldots.$$

extended over all real values of x, x', x'' etc., namely from $x = -\infty$ to $x = +\infty$, from $x' = -\infty$ to $x' = +\infty$ etc.

14.

Thus, when the function y has been reduced to the form of a sum of terms such as

$$Ax^{\alpha}x'^{\beta}x''^{\gamma}\ldots.$$

the value of the integral $\int y\psi y.\mathrm{d}y$ extended over all values of y, or the mean value of y, will be equal to the sum of terms

$$A \times \int x^{\alpha}\varphi x.\mathrm{d}x \times \int x'^{\beta}\varphi x'.\mathrm{d}x' \times \int x''^{\gamma}\varphi x''.\mathrm{d}x''\ldots.$$

where the integrations must be extended from $x = -\infty$ to $x = +\infty$, from $x' = -\infty$ to $x' = +\infty$ etc.; or, what amounts to the same thing, to the sum of the terms that arise when the individual powers $x^{\alpha}, x'^{\beta}, x''^{\gamma}$ etc. are replaced by their mean values. The truth of this very important theorem could also easily have been derived from other considerations.

15.

Let us apply what we set out in the preceding article to the special case in which

$$y = \frac{xx + x'x' + x''x'' + \text{etc.}}{\sigma}$$

where σ denotes the number of terms in the numerator. The mean value of y is immediately found here $= mm$, taking the symbol m in the same sense as above. The actual value of y in a definite case may indeed turn out greater or less than the mean, just as may the actual value of the single term xx: but the probability that the random value of y will not differ appreciably from the mean mm will approach certainty more and more as the number σ increases. To make this clearer, since it is not in our power to determine the probability itself exactly, let us investigate the mean error to be feared when we suppose $y = mm$. Evidently, by the principles of art. 6, this error will be the square root of the mean value of the function

$$\left(\frac{xx + x'x' + x''x'' + \text{etc.}}{\sigma} - mm\right)^2$$

to find which it is enough to observe that the mean value of a term such as $\frac{x^4}{\sigma\sigma}$ is $= \frac{n^4}{\sigma\sigma}$ (using the symbol n in the sense of art. 11), while the mean value of a term such as $\frac{2xxx'x'}{\sigma\sigma}$ becomes $= \frac{2m^4}{\sigma\sigma}$, whence the mean value of that function is very easily deduced

$$= \frac{n^4 - m^4}{\sigma}$$

From this we learn that, if a sufficiently large supply of mutually independent random errors x, x', x'' etc. is available, an approximate value of m can be obtained from them with great certainty by the formula

$$m = \sqrt{\frac{(xx + x'x' + x''x'' + \text{etc.})}{\sigma}}$$

and that the mean error to be feared in this determination, with regard to the square mm, is

$$= \sqrt{\frac{n^4 - m^4}{\sigma}}$$

Moreover, since the latter formula involves the quantity n, if the only aim is to form some idea of the degree of precision of this determination, it will be enough to adopt some hypothesis concerning the function φ. For example, under the third hypothesis of artt. 9, 11, that error becomes $= mm\sqrt{\frac{2}{\sigma}}$. If this is less appealing, an approximate value of n^4 can be obtained from the errors themselves with the aid of the formula

$$\frac{x^4 + x'^4 + x''^4 + \text{etc.}}{\sigma}$$

In general, however, we can assert that doubling the precision in this determination requires a fourfold supply of errors, or that the weight of the determination is proportional to the number σ.

In exactly the same way, if the errors of observations involve a constant part, an approximate value of this part can be inferred from the arithmetic mean of many errors with greater confidence the greater their number. And the mean error to be feared in this determination will be expressed by

$$\sqrt{\frac{mm - kk}{\sigma}}$$

if k denotes the constant part itself and m the mean error of the observations not yet freed of the constant part, or simply by $\frac{m}{\sqrt{\sigma}}$ if m denotes the mean error of the observations freed of the constant part (see art. 8).

16.

In artt. 12—15 we supposed that the errors x, x', x'' etc. belonged to the same kind of observations, so that their individual probabilities were expressed by the same function. But it is evident of itself that the general investigation of artt. 12—14 extends just as easily to the more general case in which the probabilities of the errors x, x', x'' etc. are expressed by different functions $\varphi x, \varphi' x', \varphi'' x''$ etc., i.e. in which those errors belong to observations of different precision or uncertainty. Let us suppose that x is the error of an observation whose mean error to be feared is $= m$; and that x', x'' etc. are the errors of other observations whose mean errors to be feared are respectively m', m'' etc. Then the mean value of the sum $xx + x'x' + x''x'' +$ etc. will be $mm + m'm' + m''m'' +$ etc.. Now if it is known from elsewhere that the quantities m, m', m'' etc. are in a given ratio, say respectively proportional to the numbers $1, \mu', \mu''$ etc., the mean value of the expression

$$\frac{xx + x'x' + x''x'' + \text{etc.}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

will be $= mm$. But if we put a definite value of the same expression, according as chance supplies the errors x, x', x'' etc., equal to mm, the mean error to which this determination remains subject is found in a similar way to that in the preceding article

$$= \frac{\sqrt{(n^4 + n'^4 + n''^4 + \text{etc.} - m^4 - m'^4 - m''^4 - \text{etc.})}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

where n', n'' etc. are supposed to signify, with regard to the observations to which the errors x', x'' etc. belong, the same thing as n with regard to the first observation. If, therefore, it is permissible to suppose the numbers n, n', n'' etc. proportional to m, m', m'' etc., that mean error to be feared becomes

$$= \frac{\sqrt{(n^4 - m^4)}.\sqrt{(1 + \mu'^4 + \mu''^4 + \text{etc.})}}{1 + \mu'\mu' + \mu''\mu'' + \text{etc.}}$$

But this way of determining an approximate value of m is not the most suitable. To show this more clearly, let us consider the more general expression

$$y = \frac{xx + \alpha'x'x' + \alpha''x''x'' + \text{etc.}}{1 + \alpha'\mu'\mu' + \alpha''\mu''\mu'' + \text{etc.}}$$

whose mean value will likewise be $= mm$, however the coefficients α', α'' etc. are chosen. The mean error to be feared when we suppose a definite value of y, according as chance supplies the errors x, x', x'' etc., equal to mm is found by the principles given above

$$= \frac{\sqrt{(n^4 - m^4 + \alpha'\alpha'(n'^4 - m'^4) + \alpha''\alpha''(n''^4 - m''^4) + \text{etc.})}}{1 + \alpha'\mu'\mu' + \alpha''\mu''\mu'' + \text{etc.}}$$

To make this mean error as small as possible, we shall have to put

$$\alpha' = \frac{n^4 - m^4}{n'^4 - m'^4}.\mu'\mu'$$

$$\alpha'' = \frac{n^4 - m^4}{n''^4 - m''^4}.\mu''\mu'' \text{ etc.}$$

Evidently these values cannot be worked out unless the ratio of the quantities n, n', n'' etc. to m, m', m'' etc. is also known from elsewhere: in the absence of this exact knowledge, it at least seems safest*) to suppose the former proportional to the latter (see art. 11), whence the values result

$$\alpha' = \frac{1}{\mu'\mu'}, \quad \alpha'' = \frac{1}{\mu''\mu''} \text{ etc.}$$

i.e. the coefficients α', α'' etc. must be put equal to the relative weights of the observations to which the errors x', x'' etc. belong, taking the weight of the observation to which the error x belongs as the unit. In this way, with σ denoting as above the number of the given errors, we shall have the mean value of the expression

$$\frac{xx + \alpha' x'x' + \alpha'' x''x'' + \text{etc.}}{\sigma}$$

$= mm$, and the mean error to be feared when we adopt a value of this expression determined by chance as the true value of mm

$$\frac{\sqrt{(n^4 + \alpha'\alpha' n'^4 + \alpha''\alpha'' n''^4 + \text{etc.} - \sigma m^4)}}{\sigma}$$

and hence, if indeed it is permissible to suppose n, n', n'' etc. proportional to m, m', m'',

$$\cdot = \sqrt{\frac{n^4 - m^4}{\sigma}}$$

which formula is identical with the one we found above for the case of observations of the same kind.

*) Namely, we conceive that knowledge of the quantities μ', μ'' etc. is within our power only in the case where, by the nature of the matter, the errors x, x', x'' etc. proportional to $1, \mu', \mu''$ etc. must be regarded as equally probable, or rather where

$$\varphi x = \mu'\varphi'(\mu' x) = \mu''\varphi''(\mu'' x) \text{ etc.}$$

17.

If the value of a quantity that depends on another unknown quantity has been determined by an observation that does not have absolute precision, the value of the unknown calculated from it will also be subject to error, but nothing in this determination is left to arbitrary choice. But if *several* quantities depending on the same unknown have become known through observations that are not absolutely exact, we can obtain the value of the unknown either from any one of these observations or from some combination of several observations, which can be done in infinitely many different ways. Although the value of the unknown arising in this way always remains subject to error, a greater error will be feared in one combination, a smaller in another. The case will be similar if several quantities depending simultaneously on several unknowns have been observed: according as the number of observations is equal to, less than or greater than the number of unknowns, the problem will be determined, undetermined or more than determined (at least generally speaking), and in the third case the observations can be combined in infinitely many different ways to determine the unknowns. To select from such a variety of combinations those most suited to the purpose, i.e. those giving values of the unknowns subject to the smallest errors, is certainly by far the most important problem in the application of mathematics to natural philosophy.

In Theoria motus corporum coelestium we showed how the *most probable* values of the unknowns are to be obtained if the law of probability of the errors of observations is known; and since this law by its nature remains hypothetical in almost all cases, we applied that theory to the most plausible law, where the probability of the error x is supposed proportional to the exponential quantity e^{-hhxx}. From this arose the method long used by us in calculations, especially astronomical ones, and now indeed used by most people doing calculations under the name of the method of least squares.

Later the illustrious LAPLACE, approaching the matter in another way, showed that the same principle must still be preferred to all others, whatever the law of probability of the errors, provided only that the number of observations is very large. But for a moderate number of observations the matter remained untouched, so that, if our hypothetical law is rejected, the method of least squares must be considered preferable to others only on the ground that it is particularly suited to convenient calculation.

We therefore hope that geometers will welcome our showing, in this new treatment of the subject, that the method of least squares gives the best combination of all. Not approximately, but absolutely, whatever the law of probability of the errors, whatever the number of observations, provided only that we establish the notion of mean error according to our treatment in artt. 5 and 6, rather than according to the view of the illustrious LAPLACE.

Moreover, it is appropriate to warn here in express words that all the following investigations concern only irregular errors free of a constant part, since the complete art of observation properly requires removing all causes of constant errors with the greatest care. What aids the calculus of probabilities itself can offer to someone undertaking the treatment of observations that there is good reason to suspect are not free of constant errors, we reserve for a separate investigation to be published on another occasion.

18.

PROBLEM. *Let U denote a given function of the unknown quantities V, V', V'' etc.; the mean error M to be feared in determining the value of U is required, if for V, V', V'' etc. we adopt their observed values rather than their true values, the observations being mutually independent and subject respectively to the mean errors m, m', m'' etc.*

Solution. Denoting the errors in the observed values of V, V', V'' etc. by e, e', e'' etc., the error thereby arising in the value of U can be expressed by the linear function

$$\lambda e + \lambda' e' + \lambda'' e'' + \text{etc.} = E$$

where $\lambda, \lambda', \lambda''$ etc. are the values of the differential quotients $\frac{\mathrm{d}U}{\mathrm{d}V}, \frac{\mathrm{d}U}{\mathrm{d}V'}, \frac{\mathrm{d}U}{\mathrm{d}V''}$ etc. for the true values of V, V', V'' etc., provided the observations are sufficiently exact to allow the squares and products of the errors to be neglected. From this it follows first, since the errors of the observations are supposed free of constant parts, that the mean value of E is $= 0$. Further, the mean error to be feared in the value of U will be the square root of the mean value of EE, or MM will be the mean value of the sum

$$\begin{aligned} &\lambda\lambda ee + \lambda'\lambda' e'e' + \lambda''\lambda'' e''e'' + \text{etc.} \\ &+ 2\lambda\lambda' ee' + 2\lambda\lambda'' ee'' + 2\lambda'\lambda'' e'e'' + \text{etc.} \end{aligned}$$

But the mean value of $\lambda\lambda ee$ becomes $\lambda\lambda mm$, the mean value of $\lambda'\lambda' e'e'$ becomes $= \lambda'\lambda' m'm'$ etc.; finally, the mean values of the products $2\lambda\lambda' ee'$ etc. all become $= 0$. Hence we conclude

$$M = \sqrt{(\lambda\lambda mm + \lambda'\lambda' m'm' + \lambda''\lambda'' m''m'' + \text{etc.})}$$

It will be appropriate to add some remarks to this solution.

I. Insofar as, by regarding the errors of observations as quantities of the first order, quantities of higher orders are neglected, we may also adopt in our formula for $\lambda, \lambda', \lambda''$ etc. those values of the quotients $\frac{\mathrm{d}U}{\mathrm{d}V}$ etc. that arise from the observed values of the quantities V, V', V'' etc. Whenever U is a linear function, there will evidently be no difference at all.

II. If we prefer to introduce the weights of the observations instead of their mean errors, let these, according to an arbitrary unit, be respectively p, p', p'' etc., and let P be the weight of the determination of the value of U arising from the observed values of the quantities V, V', V'' etc. We shall thus have

$$P = \frac{1}{\dfrac{\lambda\lambda}{p} + \dfrac{\lambda'\lambda'}{p'} + \dfrac{\lambda''\lambda''}{p''} + \text{etc.}}$$

III. If T is another given function of the quantities V, V', V'' etc., and if, for their true values,

$$\frac{\mathrm{d}T}{\mathrm{d}V} = \kappa, \quad \frac{\mathrm{d}T}{\mathrm{d}V'} = \kappa', \quad \frac{\mathrm{d}T}{\mathrm{d}V''} = \kappa'' \text{ etc.}$$

the error in the determination of the value of T obtained from the observed values of V, V', V'' etc. will be $= \kappa e + \kappa' e' + \kappa'' e'' + \text{etc.}$, $= E'$, and the mean error to be feared in this determination $= \sqrt{(\kappa\kappa mm + \kappa'\kappa' m'm' + \kappa''\kappa'' m''m'' + \text{etc.})}$. But the errors E, E' will evidently no longer be independent of one another. The mean value at issue is that of the product EE'. Unlike the mean value of the product ee', it will not be $= 0$, but $= \kappa\lambda mm + \kappa'\lambda' m'm' + \kappa''\lambda'' m''m'' + \text{etc.}$

IV. Our problem can also be extended to the case where the values of the quantities V, V', V'' etc. are not found directly by observations but are derived in any way from combinations of observations, provided only that their individual determinations are mutually independent, i.e. based on different observations: whenever this condition does not hold, the formula for M would become erroneous. For example, if one or another observation that served to determine the value of V had also been used to determine the value of V', the errors e and e' would no longer be mutually independent, nor therefore would the mean value of the product ee' be $= 0$. But if in such a case the connection of the quantities V, V' with the individual observations from which they have been derived is duly considered, the mean value of the product ee' can be assigned with the aid of remark III., and the formula for M can thus be made complete.

19.

Let V, V', V'' etc. be functions of the unknowns x, y, z etc., the number of the former $= \pi$, the number of the unknowns $= \rho$; and let us suppose that the values of the functions have been found by observations, either directly or indirectly, as $V = L, V' = L', V'' = L''$ etc., yet in such a way that these determinations were mutually independent. If ρ is greater than π, finding the unknowns evidently becomes an undetermined problem; the next case is where ρ is equal to π. Each of x, y, z etc. can be reduced, or conceived as reduced, to the form of functions of V, V', V'' etc., so that the values of the former can be found from the observed values of the latter, and at the same time, with the aid of the preceding article, the relative precision to be assigned to each of these determinations can be specified; finally, if ρ is less than π, each of x, y, z etc. can be reduced in infinitely many different ways to the form of functions of V, V', V'' etc., and thus their values can be obtained in infinitely many different ways. These determinations should indeed agree exactly if the observations had absolute precision; since the case is otherwise,

different methods will give different values, and likewise the determinations obtained from different combinations will have unequal precision.

Moreover, if in the second or third case the functions V, V', V'' etc. were such that $\pi - \rho + 1$ of them, or more, could be regarded as functions of the remaining ones, the problem would still be more than determined with regard to the latter functions, but undetermined with regard to the unknowns x, y, z etc.; their values could not be determined even if absolutely exact values of the functions V, V', V'' etc. were given: but we shall exclude this case from our investigation.

Whenever V, V', V'' etc. are not themselves *linear* functions of their indeterminates, this will be achieved if, in place of the original unknowns, we introduce their differences from approximate values. These may be supposed known from elsewhere. We shall denote the mean errors to be feared in the determinations $V = L, V' = L', V'' = L''$ etc. respectively by m, m', m'' etc., and the weights of the determinations by p, p', p'' etc., so that $pmm = p'm'm' = p''m''m''$ etc. We shall suppose the ratio of the mean errors to one another known, so that the weights, one of which can be chosen at will, are known. Finally, we shall put

$$(V - L)\sqrt{p} = v, \quad (V' - L')\sqrt{p'} = v',$$
$$(V'' - L'')\sqrt{p''} = v'' \text{ etc.}$$

Evidently, therefore, the case will be the same as if direct observations of equal precision, namely whose mean error is $= m\sqrt{p} = m'\sqrt{p'} = m''\sqrt{p''}$ etc., or to which the weight $= 1$ is assigned, had given

$$v = 0, \quad v' = 0, \quad v'' = 0 \text{ etc.}$$

20.

PROBLEM. *Let v, v', v'' etc. denote the following linear functions of the indeterminates x, y, z etc.*

$$\left.\begin{aligned} v &= ax + by + cz + \text{etc.} + l \\ v' &= a'x + b'y + c'z + \text{etc.} + l' \\ v'' &= a''x + b''y + c''z + \text{etc.} + l'' \text{ etc.} \end{aligned}\right\} \quad \text{(I)}$$

From all systems of coefficients $\kappa, \kappa', \kappa''$ etc. that give identically

$$\kappa v + \kappa' v' + \kappa'' v'' + \text{etc.} = x - k$$

with k a determinate quantity, i.e. independent of x, y, z etc., find the one for which $\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + $ etc. attains its smallest value.

Solution. Let us put

$$\left.\begin{aligned} av + a'v' + a''v'' + \text{etc.} &= \xi \\ bv + b'v' + b''v'' + \text{etc.} &= \eta \\ cv + c'v' + c''v'' + \text{etc.} &= \zeta \end{aligned}\right\} \quad \text{(II)}$$

etc.: then ξ, η, ζ etc. will also be linear functions of x, y, z etc., namely

$$\left.\begin{aligned} \xi &= x\Sigma aa + y\Sigma ab + z\Sigma ac + \text{etc.} + \Sigma al \\ \eta &= x\Sigma ab + y\Sigma bb + z\Sigma bc + \text{etc.} + \Sigma bl \\ \zeta &= x\Sigma ac + y\Sigma bc + z\Sigma cc + \text{etc.} + \Sigma cl \text{ etc.} \end{aligned}\right\} \quad \text{(III)}$$

(where Σaa denotes the sum $aa + a'a' + a''a'' + $ etc., and similarly for the others), and the number of ξ, η, ζ etc. is equal to the number of the indeterminates x, y, z etc., namely $= \rho$. By elimination, therefore, an equation of the following kind can be obtained[*)]

$$x = A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.}$$

in which, on substituting for ξ, η, ζ etc. their values from III, an identical equation must result. Thus, putting

$$\left.\begin{aligned} a[\alpha\alpha] + b[\alpha\beta] + c[\alpha\gamma] + \text{etc.} &= \alpha \\ a'[\alpha\alpha] + b'[\alpha\beta] + c'[\alpha\gamma] + \text{etc.} &= \alpha' \\ a''[\alpha\alpha] + b''[\alpha\beta] + c''[\alpha\gamma] + \text{etc.} &= \alpha'' \text{ etc.} \end{aligned}\right\} \quad \text{(IV)}$$

we shall necessarily have, identically,

$$\alpha v + \alpha' v' + \alpha'' v'' + \text{etc.} = x - A \quad \text{(V)}$$

[*)]The reason why we chose precisely these characters to denote the coefficients arising from such an elimination will become clear below.

This equation shows that the systems of values of the coefficients $\kappa, \kappa', \kappa''$ etc. must certainly also include $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc., and that, for any system, we must have identically

$$(\kappa - \alpha)v + (\kappa' - \alpha')v' + (\kappa'' - \alpha'')v'' + \text{etc.} = A - k$$

which equation implies the following

$$\begin{aligned}
(\kappa - \alpha)a + (\kappa' - \alpha')a' + (\kappa'' - \alpha'')a'' + \text{etc.} &= 0 \\
(\kappa - \alpha)b + (\kappa' - \alpha')b' + (\kappa'' - \alpha'')b'' + \text{etc.} &= 0 \\
(\kappa - \alpha)c + (\kappa' - \alpha')c' + (\kappa'' - \alpha'')c'' + \text{etc.} &= 0 \text{ etc.}
\end{aligned}$$

Multiplying these equations respectively by $[\alpha\alpha], [\alpha\beta], [\alpha\gamma]$ etc., and adding, we obtain by (IV)

$$(\kappa - \alpha)\alpha + (\kappa' - \alpha')\alpha' + (\kappa'' - \alpha'')\alpha'' + \text{etc.} = 0$$

or, what is the same,

$$\begin{aligned}
&\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.} \\
&= \alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.} \\
&\quad + (\kappa - \alpha)^2 + (\kappa' - \alpha')^2 + (\kappa'' - \alpha'')^2 + \text{etc.}
\end{aligned}$$

whence it is evident that the sum $\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.}$ attains its smallest value if we put $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc. Q. E. I.

Moreover, this smallest value itself is found in the following way. Equation (V) shows that

$$\begin{aligned}
\alpha a + \alpha' a' + \alpha'' a'' + \text{etc.} &= 1 \\
\alpha b + \alpha' b' + \alpha'' b'' + \text{etc.} &= 0 \\
\alpha c + \alpha' c' + \alpha'' c'' + \text{etc.} &= 0 \text{ etc.}
\end{aligned}$$

Multiplying these equations respectively by $[\alpha\alpha], [\alpha\beta], [\alpha\gamma]$ etc. and adding, we immediately obtain with the aid of equations (IV)

$$\alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.} = [\alpha\alpha]$$

21.

Since the observations have supplied the equations (approximately true) $v = 0, v' = 0, v'' = 0$ etc., to derive the value of the unknown x from them we must use a combination of those equations of the form

$$\kappa v + \kappa' v' + \kappa'' v'' + \text{etc.} = 0$$

which gives x the coefficient 1 and eliminates the remaining unknowns y, z etc.; by art. 18 this determination must be assigned the weight

$$= \frac{1}{\kappa\kappa + \kappa'\kappa' + \kappa''\kappa'' + \text{etc.}}$$

It therefore follows from the preceding article that the most suitable determination will be the one in which we put $\kappa = \alpha, \kappa' = \alpha', \kappa'' = \alpha''$ etc. In this way x receives the value A, and evidently the same value can also (without knowing the multipliers $\alpha, \alpha', \alpha''$ etc.) be derived directly by elimination from the equations $\xi = 0, \eta = 0, \zeta = 0$ etc. The weight to be assigned to this determination will be $= \frac{1}{[\alpha\alpha]}$, or the mean error to be feared in it

$$= m\sqrt{p}[\alpha\alpha] = m'\sqrt{p'}[\alpha\alpha] = m''\sqrt{p''}[\alpha\alpha] \text{ etc.}$$

In precisely the same way, the most suitable determination of the remaining unknowns y, z etc. will give them the same values as result from elimination from the same equations $\xi = 0, \eta = 0, \zeta = 0$ etc.

Denoting the indefinite sum $vv + v'v' + v''v''$ etc., or, what amounts to the same thing, this sum

$$p(V - L)^2 + p'(V' - L')^2 + p''(V'' - L'')^2 + \text{etc.}$$

by Ω, we see that $2\xi, 2\eta, 2\zeta$ etc. are the partial differential quotients of the function Ω, namely

$$2\xi = \frac{\mathrm{d}\Omega}{\mathrm{d}x}, \quad 2\eta = \frac{\mathrm{d}\Omega}{\mathrm{d}y}, \quad 2\zeta = \frac{\mathrm{d}\Omega}{\mathrm{d}z} \text{ etc.}$$

The values of the unknowns resulting from the most suitable combination of observations, which we may conveniently call the *most plausible values*, will therefore be identical with those for which Ω attains its minimum value. Now $V - L$ expresses indefinitely the difference between the computed and observed value. Thus the most plausible values of the unknowns will be those which make the sum of the squares of the differences between the observed and computed values of the quantities V, V', V'' etc., multiplied by the weights of the observations, a minimum: a principle that we had established by a very different route in the *Theoria Motus Corporum Coelestium*. And if the relative

precision of the individual determinations is also to be assigned, we must, by indefinite elimination from equations (III), express x, y, z etc. themselves in the following form:

$$\left.\begin{aligned} x &= A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.} \\ y &= B + [\beta\alpha]\xi + [\beta\beta]\eta + [\beta\gamma]\zeta + \text{etc.} \\ z &= C + [\gamma\alpha]\xi + [\gamma\beta]\eta + [\gamma\gamma]\zeta + \text{etc.} \end{aligned}\right\} \quad \text{(VII)}$$

$$\text{etc.}$$

In this way the most plausible values of the unknowns x, y, z etc. will be respectively A, B, C etc., and the weights to be assigned to these determinations $\frac{1}{[\alpha\alpha]}, \frac{1}{[\beta\beta]}, \frac{1}{[\gamma\gamma]}$ etc., or the mean errors to be feared in them

$$\begin{aligned} &\text{for } x \ldots\ldots m\sqrt{p}[\alpha\alpha] = m'\sqrt{p'}[\alpha\alpha] = m''\sqrt{p''}[\alpha\alpha] \text{ etc.} \\ &\text{for } y \ldots\ldots m\sqrt{p}[\beta\beta] = m'\sqrt{p'}[\beta\beta] = m''\sqrt{p''}[\beta\beta] \text{ etc.} \\ &\text{for } z \ldots\ldots m\sqrt{p}[\gamma\gamma] = m'\sqrt{p'}[\gamma\gamma] = m''\sqrt{p''}[\gamma\gamma] \text{ etc.} \end{aligned}$$

$$\text{etc.}$$

in agreement with what we taught in the *Theoria Motus Corporum Coelestium*.

22.

It will be useful to discuss separately, in a few words, the simplest and at the same time most frequent case, in which there is only one unknown and $V = x, V' = x, V'' = x$ etc. We shall then have $a = \sqrt{p}, a' = \sqrt{p'}, a'' = \sqrt{p''}$ etc., $l = -L\sqrt{p}, l' = -L'\sqrt{p'}, l'' = -L''\sqrt{p''}$ etc., and hence

$$\xi = (p + p' + p'' + \text{etc.})x - (pL + p'L' + p''L'' + \text{etc.})$$

From this, further,

$$[\alpha\alpha] = \frac{1}{p + p' + p'' + \text{etc.}}$$

$$A = \frac{pL + p'L' + p''L'' + \text{etc.}}{p + p' + p'' + \text{etc.}}$$

If, therefore, several observations of unequal precision, whose weights are respectively p, p', p'' etc., have given the value of the same quantity from the first $= L$, from the second $= L'$, from the third $= L''$ etc., its most plausible value will be

$$= \frac{pL + p'L' + p''L'' + \text{etc.}}{p + p' + p'' + \text{etc.}}$$

and the weight of this determination $= p+p'+p''$ etc. If all observations have equal precision, the most plausible value will be

$$= \frac{L + L' + L'' + \text{etc.}}{\pi}$$

that is, equal to the arithmetic mean of the observed values, and the weight of this determination $= \pi$, taking the weight of the observations as the unit.

THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS

Second Part

BY

CARL FRIEDRICH GAUSS

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THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS.

Second Part.

23.

Several investigations still remain by which the preceding theory will be both illustrated and extended.

First of all, we must investigate whether the elimination by means of which the indeterminates x, y, z etc. are to be expressed through ξ, η, ζ etc. is always possible. Since the number of the former equals that of the latter, the theory of elimination in linear equations gives the following result about this elimination. If ξ, η, ζ etc. are mutually independent, it will certainly be possible; otherwise, impossible. Let us suppose for the moment that ξ, η, ζ etc. are not mutually independent. Suppose instead that an identity exists between them

$$0 = F\xi + G\eta + H\zeta + \text{etc.} + K$$

We shall therefore have

$$\begin{aligned} F\Sigma aa + G\Sigma ab + H\Sigma ac + \text{etc.} &= 0 \\ F\Sigma ab + G\Sigma bb + H\Sigma bc + \text{etc.} &= 0 \\ F\Sigma ac + G\Sigma bc + H\Sigma cc + \text{etc.} &= 0 \end{aligned}$$

etc., and also

$$F\Sigma al + G\Sigma bl + H\Sigma cl + \text{etc.} = -K$$

Putting, further,

$$\left.\begin{aligned} aF + bG + cH + \text{etc.} &= \theta \\ a'F + b'G + c'H + \text{etc.} &= \theta' \\ a''F + b''G + c''H + \text{etc.} &= \theta'' \end{aligned}\right\} \quad \text{(I)}$$

etc., we obtain

$$\begin{aligned} a\theta + a'\theta' + a''\theta'' + \text{etc.} &= 0 \\ b\theta + b'\theta' + b''\theta'' + \text{etc.} &= 0 \\ c\theta + c'\theta' + c''\theta'' + \text{etc.} &= 0 \end{aligned}$$

etc., and also

$$l\theta + l'\theta' + l''\theta'' + \text{etc.} = -K$$

Multiplying equations (I) respectively by $\theta, \theta', \theta''$ etc. and adding, we therefore obtain:

$$0 = \theta\theta + \theta'\theta' + \theta''\theta'' + \text{etc.}$$

This equation evidently cannot hold unless $\theta = 0, \theta' = 0, \theta'' = 0$ etc. hold simultaneously. From this we first conclude that necessarily $K = 0$. Equations (I) then show that the functions v, v', v'' etc. are such that their values do not change if the values of the quantities x, y, z etc. receive increments or decrements respectively proportional to F, G, H etc.; evidently the same will hold for the functions V, V', V'' etc. Our supposition can therefore hold only in a case in which even from the exact values of the quantities V, V', V'' etc. it would have been impossible to determine the values of the unknowns x, y, z etc., that is, in which the problem would by its nature have been indeterminate, a case we excluded from our investigation.

24.

Let β, β', β'' etc. denote multipliers which have the same relation to the indeterminate y as $\alpha, \alpha', \alpha''$ etc. have to x; namely, let

$$\begin{aligned} a[\beta\alpha] + b[\beta\beta] + c[\beta\gamma] + \text{etc.} &= \beta \\ a'[\beta\alpha] + b'[\beta\beta] + c'[\beta\gamma] + \text{etc.} &= \beta' \\ a''[\beta\alpha] + b''[\beta\beta] + c''[\beta\gamma] + \text{etc.} &= \beta'' \end{aligned}$$

etc., so that identically

$$\beta v + \beta' v' + \beta'' v'' + \text{etc.} = y - B$$

Similarly, let $\gamma, \gamma', \gamma''$ etc. be analogous multipliers with respect to the indeterminate z, namely

$$\begin{aligned} a[\gamma\alpha] + b[\gamma\beta] + c[\gamma\gamma] + \text{etc.} &= \gamma \\ a'[\gamma\alpha] + b'[\gamma\beta] + c'[\gamma\gamma] + \text{etc.} &= \gamma' \\ a''[\gamma\alpha] + b''[\gamma\beta] + c''[\gamma\gamma] + \text{etc.} &= \gamma'' \end{aligned}$$

etc., so that identically

$$\gamma v + \gamma' v' + \gamma'' v'' + \text{etc.} = z - C$$

and so on. In this way, just as we already found in art. 20 that

$$\Sigma\alpha a = 1, \quad \Sigma\alpha b = 0, \quad \Sigma\alpha c = 0, \quad \text{etc., and also} \quad \Sigma\alpha l = -A$$

we shall also have

$$\begin{aligned} &\Sigma\beta a = 0, \quad \Sigma\beta b = 1, \quad \Sigma\beta c = 0, \quad \text{etc., and} \quad \Sigma\beta l = -B \\ &\Sigma\gamma a = 0, \quad \Sigma\gamma b = 0, \quad \Sigma\gamma c = 1, \quad \text{etc., and} \quad \Sigma\gamma l = -C \end{aligned}$$

and so on. Likewise, just as art. 20 gave $\Sigma\alpha\alpha = [\alpha\alpha]$, we shall also have

$$\Sigma\beta\beta = [\beta\beta], \quad \Sigma\gamma\gamma = [\gamma\gamma] \quad \text{etc.}$$

Multiplying, further, the values of $\alpha, \alpha', \alpha''$ etc. (art. 20. IV) respectively by β, β', β'' etc. and adding, we obtain

$$\alpha\beta + \alpha'\beta' + \alpha''\beta'' \quad \text{etc.} = [\alpha\beta], \quad \text{or} \quad \Sigma\alpha\beta = [\alpha\beta]$$

But multiplying the values of β, β', β'' etc. respectively by $\alpha, \alpha', \alpha''$ etc. and adding gives, in the same way,

$$\alpha\beta + \alpha'\beta' + \alpha''\beta'' + \text{etc.} = [\beta\alpha], \quad \text{hence} \quad [\alpha\beta] = [\beta\alpha]$$

In precisely the same way we obtain

$$[\alpha\gamma] = [\gamma\alpha] = \Sigma\alpha\gamma, \quad [\beta\gamma] = [\gamma\beta] = \Sigma\beta\gamma \quad \text{etc.}$$

25.

Let $\lambda, \lambda', \lambda''$ etc. further denote the values of the functions v, v', v'' etc. obtained when for x, y, z etc. we substitute their most plausible values A, B, C etc., namely

$$aA + bB + cC + \text{etc.} + l = \lambda$$
$$a'A + b'B + c'C + \text{etc.} + l' = \lambda'$$
$$a''A + b''B + c''C + \text{etc.} + l'' = \lambda''$$

etc.; let us put, in addition,

$$\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.} = M$$

so that M is the value of the function Ω corresponding to the most plausible values of the indeterminates, and hence, by what we proved in art. 20, the minimum value of this function. Then $a\lambda + a'\lambda' + a''\lambda'' + \text{etc.}$ will be the value of ξ corresponding to the values $x = A, y = B, z = C$ etc., and therefore $= 0$, that is, we shall have

$$\Sigma a\lambda = 0$$

and similarly

$$\Sigma b\lambda = 0, \quad \Sigma c\lambda = 0 \quad \text{etc.};$$
$$\text{and also} \quad \Sigma\alpha\lambda = 0, \quad \Sigma\beta\lambda = 0, \quad \Sigma\gamma\lambda = 0 \quad \text{etc.}$$

Finally, multiplying the expressions for $\lambda, \lambda', \lambda''$ etc. respectively by $\lambda, \lambda', \lambda''$ etc. and adding, we obtain $l\lambda + l'\lambda' + l''\lambda'' + \text{etc.} = \lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.}$, or

$$\Sigma l\lambda = M$$

26.

In the equation $v = ax + by + cz + \text{etc.} + l$, substituting expressions VII of art. 21 for x, y, z etc. and applying the reductions evident from what precedes gives

$$v = \alpha\xi + \beta\eta + \gamma\zeta + \text{etc.} + \lambda$$

and in the same way we shall have identically

$$v' = \alpha'\xi + \beta'\eta + \gamma'\zeta + \text{etc.} + \lambda'$$
$$v'' = \alpha''\xi + \beta''\eta + \gamma''\zeta + \text{etc.} + \lambda''$$

etc. Multiplying either these equations or equations I of art. 20 respectively by $\lambda, \lambda', \lambda''$ etc. and adding shows that identically

$$\lambda v + \lambda' v' + \lambda'' v'' + \text{etc.} = M$$

27.

The function Ω can be expressed in indefinitely many forms, which it will be worth developing. First, by squaring equations I of art. 20 and adding, we obtain immediately

$$\begin{aligned}\Omega = {} & xx\Sigma aa + yy\Sigma bb + zz\Sigma cc + \text{etc.} \\ & + 2xy\Sigma ab + 2xz\Sigma ac + 2yz\Sigma bc + \text{etc.} \\ & + 2x\Sigma al + 2y\Sigma bl + 2z\Sigma cl + \text{etc.} + \Sigma ll\end{aligned}$$

which is the *first* form.

Multiplying the same equations respectively by v, v', v'' etc. and adding, we obtain:

$$\Omega = \xi x + \eta y + \zeta z + \text{etc.} + lv + l'v' + l''v'' + \text{etc.}$$

and hence, substituting for v, v', v'' etc. the expressions given in the preceding article,

$$\Omega = \xi x + \eta y + \zeta z + \text{etc.} - A\xi - B\eta - C\zeta - \text{etc.} + M$$

or

$$\Omega = \xi(x - A) + \eta(y - B) + \zeta(z - C) + \text{etc.} + M$$

which is the *second* form.

Substituting expressions VII of art. 21 for $x - A, y - B, z - C$ etc. in the second form, we obtain the *third* form:

$$\begin{aligned}\Omega = {} & [\alpha\alpha]\xi\xi + [\beta\beta]\eta\eta + [\gamma\gamma]\zeta\zeta + \text{etc.} \\ & + 2[\alpha\beta]\xi\eta + 2[\alpha\gamma]\xi\zeta + 2[\beta\gamma]\eta\zeta + \text{etc.} + M\end{aligned}$$

To these we may add the *fourth* form, which follows of itself from the third form and the formulas of the preceding article:

$$\Omega = (v - \lambda)^2 + (v' - \lambda')^2 + (v'' - \lambda'')^2 + \text{etc.} + M, \quad \text{or}$$

$$\Omega = M + \Sigma(v - \lambda)^2$$

This form puts the condition for a minimum directly before our eyes.

28.

Let e, e', e'' etc. be the errors committed in the observations that gave $V = L, V' = L' \quad V'' = L''$ etc.; that is, let the true values of the functions V, V', V'' etc. be respectively $L - e, L' - e', L'' - e''$ etc., and hence the true values of v, v', v'' etc. respectively $-e\sqrt{p}, -e'\sqrt{p'}, -e''\sqrt{p''}$ etc. The true value of x will therefore be

$$= A - \alpha e\sqrt{p} - \alpha' e'\sqrt{p'} - \alpha'' e''\sqrt{p''} - \text{etc.}$$

or, for the error committed in the best determination of the value of x, which it is convenient to denote by Ex, we have

$$= \alpha e\sqrt{p} + \alpha' e'\sqrt{p'} + \alpha'' e''\sqrt{p''} + \text{etc.}$$

Similarly, the error committed in the best determination of the value of y, which we shall denote by Ey, will be

$$= \beta e\sqrt{p} + \beta' e'\sqrt{p'} + \beta'' e''\sqrt{p''} + \text{etc.}$$

The mean value of the square $(Ex)^2$ is found to be

$$= mmp(\alpha\alpha + \alpha'\alpha' + \alpha''\alpha'' + \text{etc.}) = mmp[\alpha\alpha]$$

and for the mean value of the square $(Ey)^2$ we similarly find $= mmp[\beta\beta]$ etc., as we have already shown above. We may now also assign the mean value of the *product* $Ex.Ey$, which is found to be

$$= mmp(\alpha\beta + \alpha'\beta' + \alpha''\beta'' + \text{etc.}) = mmp[\alpha\beta]$$

These statements can also be expressed neatly as follows. The mean values of the squares $(Ex)^2, (Ey)^2$ etc. are respectively equal to the products of $\frac{1}{2}mmp$ and the second partial derivatives

$$\frac{dd\Omega}{d\xi^2}, \quad \frac{dd\Omega}{d\eta^2} \quad \text{etc.}$$

and the mean value of a product such as $Ex.Ey$ is equal to the product of $\frac{1}{2}mmp$ and the derivative $\frac{dd\Omega}{d\xi.d\eta}$: provided, of course, that Ω is considered as a function of the indeterminates ξ, η, ζ etc.

29.

Let t denote a given linear function of the quantities x, y, z etc.; say,

$$t = fx + gy + hz + \text{etc.} + k$$

The value of t obtained from the most plausible values of x, y, z etc. will therefore be $= fA + gB + hC + \text{etc.} + k$, which we shall denote by K. If this is adopted as the true value of t, an error is committed, which will be

$$= fEx + gEy + hEz + \text{etc.}$$

and will be denoted by Et. Clearly, the mean value of this error is $= 0$, or the error will be free of a constant part. But the mean value of the square $(Et)^2$, or the mean value of the sum

$$\begin{array}{llll} ff(Ex)^2 & +2fg\,Ex.Ey & +2fh\,Ex.Ez & +\text{etc.} \\ & +gg(Ey)^2 & +2gh\,Ey.Ez & +\text{etc.} \\ & & +hh(Ez)^2 & +\text{etc. etc.} \end{array}$$

is, by what we set out in the preceding article, equal to the product of mmp and the sum

$$\begin{array}{llll} ff[\alpha\alpha] & +2fg[\alpha\beta] & +2fh[\alpha\gamma] & +\text{etc.} \\ & +gg[\beta\beta] & +2gh[\beta\gamma] & +\text{etc.} \\ & & +hh[\gamma\gamma] & +\text{etc. etc.} \end{array}$$

or to the product of mmp and the value of the function $\Omega - M$ obtained by the substitutions

$$\xi = f, \quad \eta = g, \quad \zeta = h \quad \text{etc.}$$

Thus, denoting this particular value of the function $\Omega - M$ by ω, the mean error to be feared while we adhere to the determination $t = K$ will be $= m\sqrt{p\omega}$, or the weight of this determination will be $= \frac{1}{\omega}$.

Since we have identically

$$\Omega - M = (x - A)\xi + (y - B)\eta + (z - C)\zeta + \text{etc.}$$

it is clear that ω is also equal to the particular value of the expression

$$(x - A)f + (y - B)g + (z - C)h + \text{etc.}$$

or to the particular value of $t - K$ obtained when the indeterminates x, y, z etc. are assigned those values that correspond to the values ξ, η, ζ etc. being f, g, h etc.

Finally, we observe that if t is expressed identically as a function of ξ, η, ζ etc., its constant part must be $= K$. Thus, if identically

$$t = F\xi + G\eta + H\zeta + \text{etc.} + K, \quad \text{then} \quad \omega = fF + gG + hH + \text{etc.}$$

30.

The function Ω attains its *absolute minimum* value M, as we saw above, on putting $x = A, y = B, z = C$ etc., or $\xi = 0, \eta = 0, \zeta = 0$ etc. But if one of these quantities has already been assigned a *different* value, e.g. $x = A + \Delta$, then, as the others vary, Ω can attain a *relative minimum* value, which is clearly obtained by means of the equations

$$x = A + \Delta, \quad \frac{d\Omega}{dy} = 0, \quad \frac{d\Omega}{dz} = 0 \quad \text{etc.}$$

We must therefore have $\eta = 0, \zeta = 0$ etc., and hence the relation gives

$$x = A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.}, \quad \xi = \frac{\Delta}{[\alpha\alpha]}$$

At the same time we shall have

$$y = B + \frac{[\alpha\beta]\Delta}{[\alpha\alpha]}, \quad z = C + \frac{[\alpha\gamma]\Delta}{[\alpha\alpha]} \quad \text{etc.}$$

The relative minimum value of Ω is then $= [\alpha\alpha]\xi\xi + M = M + \frac{\Delta\Delta}{[\alpha\alpha]}$. Conversely, we conclude from this that if the value of Ω must not exceed the prescribed bound $M + \mu\mu$, the value of x must necessarily lie between the bounds $A - \mu\sqrt{[\alpha\alpha]}$ and $A + \mu\sqrt{[\alpha\alpha]}$. It deserves notice that $\mu\sqrt{[\alpha\alpha]}$ is equal to the mean error to be feared in the most plausible value of x if we put $\mu = m\sqrt{p}$, that is, if μ is equal to the mean error of observations to which the weight $= 1$ is assigned.

Let us investigate more generally the minimum value of Ω that can occur for a given value of t, where, as in the preceding article, t denotes the linear function $fx + gy + hz + \text{etc.} + k$ and its most plausible value is $= K$: let the prescribed value of t be denoted by $K + \kappa$. From the theory of maxima and minima, it is known that the solution of the problem is to be sought from the equations

$$\frac{d\Omega}{dx} = \theta\frac{dt}{dx}$$
$$\frac{d\Omega}{dy} = \theta\frac{dt}{dy}$$
$$\frac{d\Omega}{dz} = \theta\frac{dt}{dz} \quad \text{etc.}$$

or $\xi = \theta f, \eta = \theta g, \zeta = \theta h$ etc., where θ denotes a multiplier that is still undetermined. Thus, if, as in the preceding article, we put *identically*

$$t = F\xi + G\eta + H\zeta + \text{etc.} + K$$

we shall have

$$K + \kappa = \theta(fF + gG + hH + \text{etc.}) + K, \quad \text{or}$$
$$\theta = \frac{\kappa}{\omega}$$

taking ω in the same sense as in the preceding article. And since $\Omega - M$ is identically a homogeneous function of the second degree in the indeterminates ξ, η, ζ etc., it follows of itself that its value for $\xi = \theta f, \eta = \theta g, \zeta = \theta h$ etc. becomes $= \theta\theta\omega$, and hence that the minimum value that Ω can attain for $t = K + \kappa$ becomes $= M + \theta\theta\omega = M + \frac{\kappa\kappa}{\omega}$. Conversely, if Ω must not exceed a prescribed value $M + \mu\mu$, the value of t must necessarily lie between the bounds $K - \mu\sqrt{\omega}$ and $K + \mu\sqrt{\omega}$, where $\mu\sqrt{\omega}$ is equal to the mean error to be feared in the most plausible determination of t if for μ we take the mean error of observations to which the weight $= 1$ is assigned.

31.

Whenever the number of quantities x, y, z etc. is somewhat larger, the numerical determination of the values A, B, C etc. from the equations $\xi = 0$, $\eta = 0$. $\zeta = 0$ etc. by ordinary elimination becomes rather laborious. For this reason, in Theoria Motus Corporum Coelestium, art. 182, we indicated a special algorithm and explained it at length in the *Disquisitione de elementis ellipticis Palladis* (Comm. recent. Soc. Gotting Vol. I), by which this labour is made as simple as the matter permits. Namely, the function Ω is to be brought into a form such as:

$$\frac{u^0 u^0}{\mathfrak{A}^0} + \frac{u' u'}{\mathfrak{B}'} + \frac{u'' u''}{\mathfrak{C}''} + \frac{u''' u'''}{\mathfrak{D}'''} + \text{etc.} + M$$

Here the divisors $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}'', \mathfrak{D}'''$ etc. are determinate quantities, while u^0, u', u'', u''' etc. are linear functions of x, y, z etc.; the second, u', however, is free of x, the third, u'', free of x and y, the fourth, u''', free of x, y and z, and so on, so that the last, $u^{(\pi-1)}$, involves only the last of the indeterminates x, y, z etc. Finally, the coefficients by which x, y, z etc. are respectively multiplied in u^0, u', u'' etc. are respectively equal to $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}''$ etc. Once this has been done, we must put $u^0 = 0, u' = 0, u'' = 0, u''' = 0$ etc., from which the values of the unknowns x, y, z etc. will be obtained most conveniently in reverse order. It seems unnecessary to repeat here the algorithm itself by which this transformation of the function Ω is carried out.

But the indeterminate elimination, by means of which the weights of these determinations must be found, requires a much more lengthy calculation still. Indeed, the weight of the determination of the last unknown (which alone enters the last $u^{(\pi-1)}$) is easily found, from what has been proved in Theoria Motus Corporum Coelestium, to be equal to the last term in the series of divisors $\mathfrak{A}^0, \mathfrak{B}', \mathfrak{C}''$ etc. Hence several calculators, to avoid that laborious elimination when other means were lacking, resorted to repeating the algorithm of which we have spoken several times, changing the order of the quantities x, y, z etc. so that each in turn occupied the last place. We therefore hope that geometers will welcome our presenting here a new method of calculating the weights of the determinations, drawn from a deeper investigation of the subject, which seems to leave nothing further to be desired.

32.

Let us therefore put (I)

$$\begin{array}{rcccccc} u^0 & = & \mathfrak{A}^0 x & +\mathfrak{B}^0 y & +\mathfrak{C}^0 z & +\text{etc.} & +\mathfrak{L}^0 \\ u' & = & & \mathfrak{B}' y & +\mathfrak{C}' z & +\text{etc.} & +\mathfrak{L}' \\ u'' & = & & & \mathfrak{C}'' z & +\text{etc.} & +\mathfrak{L}'' \\ \text{etc.} & & & & & & \end{array}$$

Then we shall have identically

$$\begin{aligned}\tfrac{1}{2}d\Omega &= \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}\\ &= \frac{u^0 du^0}{\mathfrak{A}^0} + \frac{u' du'}{\mathfrak{B}'} + \frac{u'' du''}{\mathfrak{C}''} + \text{etc.}\\ &= u^0\left(dx + \frac{\mathfrak{B}^0}{\mathfrak{A}^0}dy + \frac{\mathfrak{C}^0}{\mathfrak{A}^0}dz + \text{etc.}\right)\\ &\qquad + u'\left(dy + \frac{\mathfrak{C}'}{\mathfrak{B}'}dz + \text{etc.}\right) + u''(dz + \text{etc.}) + \text{etc.}\end{aligned}$$

from which we conclude (II)

$$\begin{aligned}\xi &= u^0\\ \eta &= \frac{\mathfrak{B}^0}{\mathfrak{A}^0}u^0 + u'\\ \zeta &= \frac{\mathfrak{C}^0}{\mathfrak{A}^0}u^0 + \frac{\mathfrak{C}'}{\mathfrak{B}'}u' + u''\\ &\text{etc.}\end{aligned}$$

Suppose that from this the following formulas are derived (III)

$$\begin{aligned}u^0 &= \xi\\ u' &= A'\xi + \eta\\ u'' &= A''\xi + B''\eta + \zeta\\ &\text{etc.}\end{aligned}$$

Now, from the total differential of the equation

$$\Omega = \xi(x - A) + \eta(y - B) + \zeta(z - C) + \text{etc.} + M$$

subtracting the equation

$$\tfrac{1}{2}d\Omega = \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}$$

gives

$$\tfrac{1}{2}d\Omega = (x - A)d\xi + (y - B)d\eta + (z - C)d\zeta + \text{etc.}$$

This expression must be identical with the following one derived from III:

$$\begin{aligned}&\frac{u^0}{\mathfrak{A}^0}.d\xi + \frac{u'}{\mathfrak{B}'}(A'd\xi + d\eta)\\ &\qquad + \frac{u''}{\mathfrak{C}''}(A''d\xi + B''d\eta + d\zeta) + \text{etc.}\end{aligned}$$

From this we conclude (IV)

$$\begin{array}{rcccccc} x & = & \frac{u^0}{\mathfrak{A}^0} & +A'.\frac{u'}{\mathfrak{B}'} & +A''.\frac{u''}{\mathfrak{C}''} & +\text{etc.} + A \\ y & = & & \frac{u'}{\mathfrak{B}'} & +B''.\frac{u''}{\mathfrak{C}''} & +\text{etc.} + B \\ z & = & & & \frac{u''}{\mathfrak{C}''} & +\text{etc.} + C \\ \text{etc.} & & & & & & \end{array}$$

Substituting in these expressions for u^0, u', u'' etc. their values taken from III will complete the indeterminate elimination. In particular, to determine the weights we shall have (V)

$$\begin{array}{rcccc} [\alpha\alpha] = & \frac{1}{\mathfrak{A}^0} + \frac{A'A'}{\mathfrak{B}'} & + \frac{A''A''}{\mathfrak{C}''} & + \frac{A'''A'''}{\mathfrak{D}'''} & + \text{etc.} \\ [\beta\beta] = & \frac{1}{\mathfrak{B}'} + \frac{B''B''}{\mathfrak{C}''} & + \frac{B'''B'''}{\mathfrak{D}'''} & + \text{etc.} & \\ [\gamma\gamma] = & \frac{1}{\mathfrak{C}''} + \frac{C'''C'''}{\mathfrak{D}'''} & + \text{etc.} & & \\ \text{etc.} & & & & \end{array}$$

The simplicity of these formulas leaves nothing to be desired. Equally simple formulas also result for the remaining coefficients $[\alpha\beta], [\alpha\gamma], [\beta\gamma]$ etc., but since their use is less frequent, we refrain from giving them here.

33.

Because of the importance of the matter, and so that everything may be ready for calculation, it seemed appropriate also to write down here explicit formulas for determining the coefficients A', A'', A''' etc., B'', B''' etc. etc. This calculation can be carried out in two ways, since identities must result both when the values of u^0, u', u'' etc. taken from III are substituted in II, and when the values of ξ, η, ζ etc. from II are substituted in III. The first way supplies the following systems of formulas:

$$\frac{\mathfrak{B}^0}{\mathfrak{A}^0} + A' = 0$$

$$\frac{\mathfrak{C}^0}{\mathfrak{A}^0} + \frac{\mathfrak{C}'}{\mathfrak{B}'}.A' + A'' = 0$$

$$\frac{\mathfrak{D}^0}{\mathfrak{A}^0} + \frac{\mathfrak{D}'}{\mathfrak{B}'}.A' + \frac{\mathfrak{D}''}{\mathfrak{C}''}.A'' + A''' = 0$$

etc., from which A', A'', A''' etc. are found.

$$\frac{\mathfrak{C}'}{\mathfrak{B}'} + B'' = 0$$

$$\frac{\mathfrak{D}'}{\mathfrak{B}'} + \frac{\mathfrak{D}''}{\mathfrak{C}''}.B'' + B''' = 0$$

etc., from which B'', B''' etc. are found.

$$\frac{\mathfrak{D}''}{\mathfrak{C}''} + C''' = 0$$

etc., from which C''' etc. are found. And so on.

The other way gives these formulas:

$$\mathfrak{A}^0 A' + \mathfrak{B}^0 = 0$$

from which A' is obtained.

$$\begin{array}{llll} \mathfrak{A}^0 A'' & +\mathfrak{B}^0 B'' & +\mathfrak{C}^0 & = 0 \\ & \mathfrak{B}' B'' & +\mathfrak{C}' & = 0 \end{array}$$

from which B'' and A'' are found.

$$\begin{array}{llllll} \mathfrak{A}^0 A''' & +\mathfrak{B}^0 B''' & +\mathfrak{C}^0 C''' & +\mathfrak{D}^0 & = 0 \\ & \mathfrak{B}' B''' & +\mathfrak{C}' C''' & +\mathfrak{D}' & = 0 \\ & & \mathfrak{C}'' C''' & +\mathfrak{D}'' & = 0 \end{array}$$

from which C''', B''', A''' are found. And so on.

The two ways are almost equally convenient if the weights of the determinations of all the quantities x, y, z etc. are wanted; but whenever only one or two of the quantities $[\alpha\alpha], [\beta\beta], [\gamma\gamma]$ etc. are required, the first system will clearly be far preferable.

Moreover, combining equations I with IV leads to the same formulas and also supplies two ways of calculating the most plausible values A, B, C etc. themselves; namely, *first*,

$$A = -\frac{\mathfrak{L}^0}{\mathfrak{A}^0} - A'\frac{\mathfrak{L}'}{\mathfrak{B}'} - A''\frac{\mathfrak{L}''}{\mathfrak{C}''} - A'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$B = -\frac{\mathfrak{L}'}{\mathfrak{B}'} - B''\frac{\mathfrak{L}''}{\mathfrak{C}''} - B'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$C = -\frac{\mathfrak{L}''}{\mathfrak{C}''} - C'''\frac{\mathfrak{L}'''}{\mathfrak{D}'''} - \text{etc.}$$

$$\text{etc.}$$

The other calculation is identical with the ordinary one, in which we put $u^0 = 0, u' = 0, u'' = 0$ etc.

34.

What we set out in art. 32 consists only of special cases of a more general theorem, which is as follows:

THEOREM. *Let t denote the following linear function of the indeterminates x, y, z etc.*

$$t = fx + gy + hz + \text{etc.} + k$$

which, when transformed into a function of the indeterminates u^0, u', u'' etc., becomes

$$t = k^0u^0 + k'u' + k''u'' + \text{etc.} + K$$

Then K will be the most plausible value of t, and the weight of this determination will be

$$= \frac{1}{\mathfrak{A}^0k^0k^0 + \mathfrak{B}'k'k' + \mathfrak{C}''k''k'' + \text{etc.}}$$

Proof. The first part of the theorem is clear because the most plausible value of t must correspond to the values $u^0 = 0, u' = 0, u'' = 0$ etc. To prove the latter part, we observe that, since $\frac{1}{2}d\Omega = \xi\,dx + \eta\,dy + \zeta\,dz + \text{etc.}$ and $dt = f\,dx + g\,dy + h\,dz + \text{etc.}$, the following holds for $\xi = f.\quad \eta = g, \zeta = h$ etc., independently of the values of the differentials dx, dy, dz etc.:

$$d\Omega = 2dt$$

It therefore follows that, for the same values $\xi = f, \eta = g, \zeta = h$ etc.,

$$\frac{u^0}{\mathfrak{A}^0}du^0 + \frac{u'}{\mathfrak{B}'}du' + \frac{u''}{\mathfrak{C}''}.du'' + \text{etc.}$$
$$= k^0du^0 + k'du' + k''du'' + \text{etc.}$$

It is now readily seen that if dx, dy, dz etc. are independent of one another, then du^0, du', du'' etc. are also independent of one another; from this we conclude that for $\xi = f, \eta = g, \zeta = h$ etc.,

$$u^0 = \mathfrak{A}^0 k^0, \quad u' = \mathfrak{B}' k', \quad u'' = \mathfrak{C}'' k'' \quad \text{etc.}$$

The value of Ω corresponding to the same values will therefore be

$$= \mathfrak{A}^0 k^0 k^0 + \mathfrak{B}' k' k' + \mathfrak{C}'' k'' k'' + \text{etc.} + M$$

from which the truth of our theorem follows directly by art. 29.

If, moreover, we wish to carry out the transformation of the function t directly, i. e. without knowing substitutions IV of art. 32, the following formulas are available:

$$\begin{aligned} f &= \mathfrak{A}^0 k^0 \\ g &= \mathfrak{B}^0 k^0 + \mathfrak{B}' k' \\ h &= \mathfrak{C}^0 k^0 + \mathfrak{C}' k' + \mathfrak{C}'' k'' \quad \text{etc.}, \end{aligned}$$

from which the coefficients k^0, k', k'' etc. will be determined successively, and finally we shall have

$$K = k - \mathfrak{L}^0 k^0 - \mathfrak{L}' k' - \mathfrak{L}'' k'' - \text{etc.}$$

35.

The following problem deserves a separate treatment, both because of its practical usefulness and because of the elegance of its solution.

To find the changes in the most plausible values of the unknowns produced by the addition of a new equation, and the weights of the new determinations.

We shall retain the notation used above, so that the original equations, reduced to weight = 1, are these: $v = 0, v' = 0, v'' = 0$ etc.; for the indefinite sum we write $vv + v'v' + v''v''$ etc. $= \Omega$; and ξ, η, ζ etc. are the partial differential quotients

$$\frac{d\Omega}{2dx}, \quad \frac{d\Omega}{2dy}, \quad \frac{d\Omega}{2dz} \quad \text{etc.}$$

and finally indefinite elimination gives

$$\left.\begin{aligned} x &= A + [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.} \\ y &= B + [\alpha\beta]\xi + [\beta\beta]\eta + [\beta\gamma]\zeta + \text{etc.} \\ z &= C + [\alpha\gamma]\xi + [\beta\gamma]\eta + [\gamma\gamma]\zeta + \text{etc.} \end{aligned}\right\} \quad \text{(I)}$$

Now let us suppose that a new equation $v^* = 0$ is added (approximately true, and with weight = 1), and let us investigate how much this will change both the most plausible values of the unknowns A, B, C etc. and the coefficients $[\alpha\alpha], [\alpha\beta]$ etc.

Let us put $\Omega + v^*v^* = \Omega^*$, $\frac{d\Omega^*}{2dx} = \xi^*$, $\frac{d\Omega^*}{2dy} = \eta^*$, $\frac{d\Omega^*}{2dz} = \zeta^*$ etc., and suppose that elimination then gives

$$x = A^* + [\alpha\alpha^*]\xi^* + [\alpha\beta^*]\eta^* + [\alpha\gamma^*]\zeta^* \quad \text{etc.}$$

Finally, let

$$v^* = fx + gy + hz + \text{etc.} + k$$

so that, on substituting for x, y, z etc. their values from (I), we obtain

$$v^* = F\xi + G\eta + H\zeta + \text{etc.} + K$$

and let us put

$$Ff + Gg + Hh + \text{etc.} = \omega$$

Clearly K will be the most plausible value of the function v^* as far as it follows from the original equations, without regard to the value 0 supplied by the additional observation, and $\frac{1}{\omega}$ will be the weight of this determination.

We now have

$$\xi^* = \xi + fv^*, \quad \eta^* = \eta + gv^*, \quad \zeta^* = \zeta + hv^* \quad \text{etc.}$$

and therefore

$$\begin{aligned} &F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K \\ &\quad = v^*(1 + Ff + Gg + Hh + \text{etc.}) \end{aligned}$$

or

$$v^* = \frac{F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K}{1+\omega}$$

Similarly we obtain

$$\begin{aligned} x &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} \\ &\quad - v^*(f[\alpha\alpha] + g[\alpha\beta] + h[\alpha\gamma] + \text{etc.}) \\ &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} - Fv^* \\ &= A + [\alpha\alpha]\xi^* + [\alpha\beta]\eta^* + [\alpha\gamma]\zeta^* + \text{etc.} \\ &\quad - \frac{F}{1+\omega}(F\xi^* + G\eta^* + H\zeta^* + \text{etc.} + K) \end{aligned}$$

From this we therefore conclude that

$$A^* = A - \frac{FK}{1+\omega}$$

which will be the most plausible value of x from *all* the observations;

$$[\alpha\alpha^*] = [\alpha\alpha] - \frac{FF}{1+\omega}$$

and the weight of this determination is therefore

$$= \frac{1}{[\alpha\alpha] - \frac{FF}{1+\omega}}$$

In exactly the same way we find the most plausible value of y, based on *all* the observations,

$$B^* = B - \frac{GK}{1+\omega}$$

and the weight of this determination is

$$= \frac{1}{[\beta\beta] - \frac{GG}{1+\omega}}$$

and so on. Q. E. I.

Let us add a few remarks to this solution.

I. On substituting these new values A^*, B^*, C^* etc., the function v^* acquires the most plausible value

$$K - \frac{K}{1+\omega}(Ff + Gg + Hh + \text{etc.}) = \frac{K}{1+\omega}$$

And since we have indefinitely

$$v^* = \frac{F}{1+\omega}.\xi^* + \frac{G}{1+\omega}.\eta^* + \frac{H}{1+\omega}.\zeta^* + \text{etc.} + \frac{K}{1+\omega}$$

the weight of this determination is found by the principles of art. 29 to be

$$= \frac{1+\omega}{Ff+Gg+Hh+\text{etc.}} = \frac{1}{\omega}+1$$

The same results follow directly by applying the rule given at the end of art. 21; namely, the system of original equations had supplied the determination $v^* = K$ with weight $= \frac{1}{\omega}$, and the new observation then gave another determination, independent of the former, $v^* = 0$, with weight $= 1$; combining them gives the determination $v^* = \frac{K}{1+\omega}$ with weight $= \frac{1}{\omega}+1$.

II. It further follows that, since for $x = A^*, y = B^*, z = C^*$ etc. we must have $\xi^* = 0, \eta^* = 0, \zeta^* = 0$ etc., the same values give

$$\xi = -\frac{fK}{1+\omega}, \quad \eta = -\frac{gK}{1+\omega}, \quad \zeta = -\frac{hK}{1+\omega} \quad \text{etc.}$$

and also, since indefinitely $\Omega = \xi(x-A) + \eta(y-B) + \zeta(z-C) + \text{etc.} + M$,

$$\begin{aligned}\Omega &= \frac{KK}{(1+\omega)^2}(Ff+Gg+Hh+\text{etc.}) + M \\ &= M + \frac{\omega KK}{(1+\omega)^2}\end{aligned}$$

and finally, since indefinitely $\Omega^* = \Omega + v^*v^*$,

$$\Omega^* = M + \frac{\omega KK}{(1+\omega)^2} + \frac{KK}{(1+\omega)^2} = M + \frac{KK}{1+\omega}$$

III. Comparing this with what we taught in art. 30, we observe that the function Ω here attains the smallest value it can take for the specified value of the function $v^* = \frac{K}{1+\omega}$.

36.

For another problem, related to the preceding one, namely

To investigate the changes in the most plausible values of the unknowns arising from a change in the weight of one of the original observations, and the weights of the new determinations

we shall give only the solution here, omitting for brevity the proof, which is easily completed on the model of the preceding article.

Let us suppose that only after completing the calculation we notice that one of the observations has been assigned a weight either too small or too large; e. g. that the first observation, which gave $V = L$, should be assigned,

instead of the weight p used in the calculation, more correctly a weight $= p^*$. There will then be no need to repeat the entire calculation; instead, the corrections can be computed more conveniently by the following formulas.

The corrected most plausible values of the unknowns will be these:

$$x = A - \frac{(p^* - p)\alpha\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$y = B - \frac{(p^* - p)\beta\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$z = C - \frac{(p^* - p)\gamma\lambda}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

etc., and the weights of these determinations will be found by dividing unity respectively by

$$[\alpha\alpha] - \frac{(p^* - p)\alpha\alpha}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$[\beta\beta] - \frac{(p^* - p)\beta\beta}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})}$$

$$[\gamma\gamma] - \frac{(p^* - p)\gamma\gamma}{p + (p^* - p)(a\alpha + b\beta + c\gamma + \text{etc.})} \quad \text{etc.}$$

This solution also includes the case where, after completing the calculation, we realize that one of the observations should have been rejected altogether, since this is the same as putting $p^* = 0$; and similarly the value $p^* = \infty$ refers to the case where the equation $V = L$, which was treated as approximate in the calculation, actually has absolute precision.

Moreover, whenever either *several* new equations are added to those on which the calculation was based, or we realize that *several* of them have been assigned incorrect weights, the calculation of the corrections would become too complicated; in such a case it will therefore be better to redo the calculation from the beginning.

37.

In arts. 15. 16 we explained a method for determining the precision of observations approximately[*]. But this method assumes that a sufficiently large number of the errors that actually occurred are known exactly, a condition which, strictly speaking, will very rarely, not to say never, be met. If, indeed, the quantities whose approximate values have become known through observations depend, according to a known law, on one or more unknown quantities, the most plausible values of these unknowns can be found by the method of least squares. The values of the quantities that were the objects of observation, computed from those values, will then be supposed to differ very little from the true values, so that the greater the number of observations, the more justification there will be for taking their differences from the observed values as the true errors of the observations. This practice has been followed by all calculators who have undertaken to estimate the precision of observations a posteriori in particular cases: but it is clearly wrong in theory, and although in many cases it may suffice for practical purposes, in others it can lead to enormous errors. This subject therefore very much deserves a more careful examination.

In this investigation we shall retain the notation used from art. 19 onward. The practice we have described regards the quantities A, B, C etc. as the true values of x, y, z, and consequently $\lambda, \lambda', \lambda''$ etc. as the true values of the functions v, v', v'' etc. If all observations have equal precision and their weight $p = p' = p''$ etc. has been taken as unity, the same quantities, with their signs changed, represent the errors of the observations under that assumption; the rules of art. 15 then give the mean error of the observations m

$$= \sqrt{\frac{\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.}}{\pi}} = \sqrt{\frac{M}{\pi}}$$

If the observations do not have the same precision, the quantities $-\lambda, -\lambda', -\lambda''$ etc. represent the errors of the observations multiplied by the square roots of their weights, and the rules of art. 16 would lead to the same formula $\sqrt{\frac{M}{\pi}}$, now denoting the mean error of observations assigned weight $= 1$. But clearly an exact calculation would require that, instead of the quantities $\lambda, \lambda', \lambda''$ etc., the values of the functions v, v', v'' etc. obtained from the true values of x, y, z etc. be used; i. e. instead of M, the value of the function Ω corresponding to the true values of x, y, z etc. Although this value cannot be specified, we are nevertheless certain that it is greater than M (which is

[*] The investigation of the same subject which we had presented in an earlier paper (*Bestimmung der Genauigkeit der Beobachtungen. Zeitschrift für Astronomie und verwandte Wissenschaften* Vol. I, p. 185) rested on the same hypothesis concerning the nature of the function expressing the probability of errors on which we had based the method of least squares in Theoria motus corporum coelestium (see art. 9, III).

the smallest possible value), except in the infinitely improbable case where the most plausible values of the unknowns agree exactly with the true values. In general, therefore, we can assert that the usual practice produces a mean error smaller than it should be, or assigns the observations too great a precision. Let us now see what rigorous theory teaches.

38.

First of all it is necessary to investigate how M depends on the true errors of the observations. Let us denote these, as in art. 28, by e, e', e'' etc., and for greater simplicity put

$$e\sqrt{p} = \varepsilon, \quad e'\sqrt{p'} = \varepsilon', \quad e''\sqrt{p''} = \varepsilon'' \quad \text{etc., and also}$$
$$m\sqrt{p} = m'\sqrt{p'} = m''\sqrt{p''} \quad \text{etc.} = \mu$$

Further, let the true values of x, y, z etc. be, respectively, $A - x^0, B - y^0, C - z^0$ etc., and let the corresponding values of ξ, η, ζ etc. be $-\xi^0, -\eta^0, -\zeta^0$ etc. Clearly the same values will give for v, v', v'' etc. the values $-\varepsilon, -\varepsilon', -\varepsilon''$ etc., so that the following holds

$$\xi^0 = a\varepsilon + a'\varepsilon' + a''\varepsilon'' + \text{etc.}$$
$$\eta^0 = b\varepsilon + b'\varepsilon' + b''\varepsilon'' + \text{etc.}$$
$$\zeta^0 = c\varepsilon + c'\varepsilon' + c''\varepsilon'' + \text{etc.}$$

etc., and also

$$x^0 = \alpha\varepsilon + \alpha'\varepsilon' + \alpha''\varepsilon'' + \text{etc.}$$
$$y^0 = \beta\varepsilon + \beta'\varepsilon' + \beta''\varepsilon'' + \text{etc.}$$
$$z^0 = \gamma\varepsilon + \gamma'\varepsilon' + \gamma''\varepsilon'' + \text{etc.}$$

Finally, let us put

$$\Omega^0 = \varepsilon\varepsilon + \varepsilon'\varepsilon' + \varepsilon''\varepsilon'' + \text{etc.}$$

so that Ω^0 is equal to the value of the function Ω corresponding to the true values of x, y, z etc. Since the following holds indefinitely

$$\Omega = M + (x - A)\xi + (y - B)\eta + (z - C)\zeta + \text{etc.}$$

it also follows that

$$M = \Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.}$$

It is therefore clear that M, when expanded, is a homogeneous function of the second order of the errors e, e', e'' etc., which may become larger or smaller according to the different values of the errors. But insofar as the sizes of the errors remain unknown to us, it will be appropriate to consider this function

indefinitely and, above all, to assign its mean value according to the principles of the calculus of probability. We shall find this by writing, in place of the squares $ee, e'e', e''e''$ etc., respectively, $mm, m'm', m''m''$ etc., while omitting the products $ee', ee'', e'e''$ etc. altogether; or, what is the same, by writing in place of each square $\varepsilon\varepsilon, \varepsilon'\varepsilon', \varepsilon''\varepsilon''$ etc. the value $\mu\mu$, and disregarding the products $\varepsilon\varepsilon', \varepsilon\varepsilon'', \varepsilon'\varepsilon''$ etc. entirely. In this way the term Ω^0 clearly gives $\pi\mu\mu$; the term $-x^0\xi^0$ will give

$$-(a\alpha + a'\alpha' + a''\alpha'' + \text{etc.})\mu\mu = -\mu\mu$$

and similarly each of the remaining parts will give $-\mu\mu$, so that the total mean value becomes $= (\pi - \rho)\mu\mu$, where π denotes the number of observations and ρ the number of unknowns. The true value of M may indeed be greater or smaller than the mean, according to the errors chance has produced; but the greater the number of observations, the less significant the discrepancy will be, so that the following can be taken as an approximate value of μ:

$$\sqrt{\frac{M}{\pi - \rho}}$$

Thus the value of μ resulting from the erroneous practice discussed in the preceding article must be increased in the ratio of the quantity $\sqrt{(\pi - \rho)}$ to $\sqrt{\pi}$.

39.

To make clearer how far it is legitimate to equate the value of M arising by chance with its mean, it is still necessary to investigate the mean error to be feared when we put $\frac{M}{\pi-\rho} = \mu\mu$. This mean error is equal to the square root of the mean value of the quantity

$$\left(\frac{\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.} - (\pi-\rho)\mu\mu}{\pi-\rho}\right)^2$$

which we shall express as follows:

$$\left(\frac{\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.}}{\pi-\rho}\right)^2$$
$$- \frac{2\mu\mu}{\pi-\rho}(\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.} - (\pi-\rho)\mu\mu) - \mu^4$$

and since the mean value of the second term is clearly $= 0$, what we have to investigate is the mean value of the function

$$\Psi = (\Omega^0 - x^0\xi^0 - y^0\eta^0 - z^0\zeta^0 - \text{etc.})^2$$

Once this has been found and denoted by N, the required mean error will be

$$= \sqrt{\left(\frac{N}{(\pi-\rho)^2} - \mu^4\right)}$$

The expanded expression Ψ is clearly a homogeneous function either of the errors e, e', e'' etc. or of the quantities $\varepsilon, \varepsilon', \varepsilon''$ etc., and its mean value will be found if

1° the fourth powers e^4, e'^4, e''^4 etc. are replaced by their mean values

2° each product of two squares, such as $eee'e', eee''e'', e'e'e''e''$ etc., is replaced by the product of their mean values, namely $mmm'm', mmm''m'', m'm'm''m''$ etc.

3° all the remaining parts, which will involve either a factor such as e^3e' or one such as $eee'e''$, are omitted altogether.

We shall assume the mean values of the fourth powers e^4, e'^4, e''^4 etc. to be proportional to the fourth powers m^4, m'^4, m''^4 etc. themselves (see art. 16), so that the former are to the latter as ν^4 to μ^4; thus ν^4 denotes the mean value of the fourth powers of observations whose weight is $= 1$. The preceding rules can therefore also be expressed as follows: in place of each fourth power $\varepsilon^4, \varepsilon'^4, \varepsilon''^4$ etc. we must write ν^4, in place of each product of two squares, such as $\varepsilon\varepsilon\varepsilon'\varepsilon', \varepsilon\varepsilon\varepsilon''\varepsilon'', \varepsilon'\varepsilon'\varepsilon''\varepsilon''$ etc., we must write μ^4, and all remaining terms that will involve factors such as $\varepsilon^3\varepsilon'$ or $\varepsilon\varepsilon\varepsilon'\varepsilon''$ or $\varepsilon\varepsilon'\varepsilon''\varepsilon'''$ must be suppressed.

With this properly understood, it will readily be seen that

I. The mean value of the square $\Omega^0\Omega^0$ is $\pi\nu^4 + (\pi\pi - \pi)\mu^4$

II. The mean value of the product $\varepsilon\varepsilon x^0\xi^0$ is $= a\alpha\nu^4 + (a'\alpha' + a''\alpha'' + \text{etc.})\mu^4$, or, since $a\alpha + a'\alpha' + a''\alpha'' + \text{etc.} = 1$,

$$= a\alpha(\nu^4 - \mu^4) + \mu^4$$

And since, similarly, the mean value of the product $\varepsilon'\varepsilon' x^0\xi^0$ is $= a'\alpha'(\nu^4 - \mu^4) + \mu^4$, while the mean value of the product $\varepsilon''\varepsilon'' x^0\xi^0$ is $= a''\alpha''(\nu^4 - \mu^4) + \mu^4$ and so on, it is clear that the mean value of the product $(\varepsilon\varepsilon + \varepsilon'\varepsilon' + \varepsilon''\varepsilon'' + \text{etc.})x^0\xi^0$ or $\Omega^0 x^0\xi^0$ is

$$= \nu^4 - \mu^4 + \pi\mu^4$$

The products $\Omega^0 y^0\eta^0, \Omega^0 z^0\zeta^0$ etc. will have the same mean value. Therefore the mean value of the product $\Omega^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is

$$= \rho\nu^4 + \rho(\pi - 1)\mu^4$$

III. To keep the remaining expansions from becoming too lengthy, a suitable notation must be introduced. We shall therefore use the symbol Σ in a somewhat wider sense than has been done at various places above: it will denote the sum of the term to which it is prefixed and all similar but non-identical terms arising from it by all permutations of the observations. In this way, e. g., we have $x^0 = \Sigma\alpha\varepsilon$, $x^0x^0 = \Sigma\alpha\alpha\varepsilon\varepsilon + 2\Sigma\alpha\alpha'\varepsilon\varepsilon'$. Collecting the mean value of the product $x^0x^0\xi^0\xi^0$ by parts, we therefore first have the mean value of the product $\alpha\alpha\varepsilon\varepsilon\xi^0\xi^0$

$$\begin{aligned} &= aa\alpha\alpha\nu^4 + \alpha\alpha(a'a' + a''a'' + \text{etc.})\mu^4 \\ &= aa\alpha\alpha(\nu^4 - \mu^4) + \alpha\alpha\mu^4\Sigma aa \end{aligned}$$

Similarly the mean value of the product $\alpha'\alpha'\varepsilon'\varepsilon'\xi^0\xi^0$ is $= a'a'\alpha'\alpha'(\nu^4 - \mu^4) + \alpha'\alpha'\mu^4\Sigma aa$ and so on, and therefore the mean value of the product $\xi^0\xi^0\Sigma\alpha\alpha\varepsilon\varepsilon$ is

$$= (\nu^4 - \mu^4)\Sigma aa\alpha\alpha + \mu^4\Sigma aa.\Sigma\alpha\alpha$$

Further, the mean value of the product $\alpha\alpha'\varepsilon\varepsilon'\xi^0\xi^0$ is $= 2aa'\alpha\alpha'\mu^4$, and similarly the mean value of the product $\alpha\alpha''\varepsilon\varepsilon''\xi^0\xi^0$ is $= 2aa''\alpha\alpha''\mu^4$ etc.; from this it is readily concluded that the mean value of the product $\xi^0\xi^0\Sigma\alpha\alpha'\varepsilon\varepsilon'$ is

$$\begin{aligned} &= 2\mu^4\Sigma a\alpha a'\alpha' = \mu^4((\Sigma a\alpha)^2 - \Sigma aa\alpha\alpha) \\ &= \mu^4(1 - \Sigma aa\alpha\alpha) \end{aligned}$$

Collecting these, we have the mean value of the product $x^0x^0\xi^0\xi^0$

$$= (\nu^4 - 3\mu^4)\Sigma aa\alpha\alpha + 2\mu^4 + \mu^4\Sigma aa.\Sigma\alpha\alpha$$

IV. In a very similar way the mean value of the product $x^0y^0\xi^0\eta^0$ is found to be

$$\begin{aligned} &= \nu^4\Sigma ab\alpha\beta + \mu^4\Sigma a\alpha b'\beta' \\ &\quad + \mu^4\Sigma ab\alpha'\beta' + \mu^4\Sigma a\beta b'\alpha' \end{aligned}$$

But the following holds

$$\begin{aligned} \Sigma a\alpha b'\beta' &= \Sigma a\alpha.\Sigma b\beta - \Sigma a\alpha b\beta \\ \Sigma ab\alpha'\beta' &= \Sigma ab.\Sigma\alpha\beta - \Sigma ab\alpha\beta \\ \Sigma a\beta b'\alpha' &= \Sigma a\beta.\Sigma b\alpha - \Sigma a\beta b\alpha \end{aligned}$$

therefore, because $\Sigma a\alpha = 1, \Sigma b\beta = 1, \Sigma a\beta = 0, \Sigma b\alpha = 0$, that mean value becomes

$$= (\nu^4 - 3\mu^4)\Sigma ab\alpha\beta + \mu^4(1 + \Sigma ab.\Sigma\alpha\beta)$$

V. Since, in exactly the same way, the mean value of the product $x^0z^0\xi^0\zeta^0$ is

$$= (\nu^4 - 3\mu^4)\Sigma ac\alpha\gamma + \mu^4(1 + \Sigma ac.\Sigma\alpha\gamma)$$

and so on, addition gives the mean value of the product $x^0\xi^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ as

$$\begin{aligned} &= (\nu^4 - 3\mu^4)\Sigma(a\alpha(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 1)\mu^4 \\ &\quad + \mu^4(\Sigma aa.\Sigma\alpha\alpha + \Sigma ab.\Sigma\alpha\beta + \Sigma ac.\Sigma\alpha\gamma + \text{etc.}) \\ &= (\nu^4 - 3\mu^4)\Sigma(a\alpha(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4 \end{aligned}$$

VI. In exactly the same way the mean value of the product $y^0\eta^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is found to be

$$= (\nu^4 - 3\mu^4)\Sigma(b\beta(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4$$

then the mean value of the product $z^0\zeta^0(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})$ is

$$= (\nu^4 - 3\mu^4)\Sigma(c\gamma(a\alpha + b\beta + c\gamma + \text{etc.})) + (\rho + 2)\mu^4$$

and so on. Addition therefore gives the mean value of the square $(x^0\xi^0 + y^0\eta^0 + z^0\zeta^0 + \text{etc.})^2$

$$= (\nu^4 - 3\mu^4)\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2) + (\rho\rho + 2\rho)\mu^4$$

VII. Finally, collecting everything properly gives

$$\begin{aligned} N &= (\pi - 2\rho)\nu^4 + (\pi\pi - \pi - 2\pi\rho + 4\rho + \rho\rho)\mu^4 \\ &\quad + (\nu^4 - 3\mu^4)\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2) \\ &= (\pi - \rho)(\nu^4 - \mu^4) + (\pi - \rho)^2\mu^4 \\ &\quad - (\nu^4 - 3\mu^4)[\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)] \end{aligned}$$

Thus the mean error to be feared in determining $\mu\mu$ by the formula

$$\mu\mu = \frac{M}{\pi - \rho}$$

will be

$$= \sqrt{\left\{ \frac{\nu^4 - \mu^4}{\pi - \rho} - \frac{\nu^4 - 3\mu^4}{(\pi - \rho)^2} . [\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)] \right\}}$$

40.

The quantity $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$, which enters the expression just found, cannot in general be reduced to a simpler form. Nevertheless two bounds can be assigned between which its value must necessarily lie. *First*, from the relations developed above it is readily proved that

$$\begin{aligned} &(a\alpha + b\beta + c\gamma + \text{etc.})^2 + (a\alpha' + b\beta' + c\gamma' + \text{etc.})^2 \\ &\quad + (a\alpha'' + b\beta'' + c\gamma'' + \text{etc.})^2 + \text{etc.} \\ &= a\alpha + b\beta + c\gamma + \text{etc.} \end{aligned}$$

from which we conclude that $a\alpha + b\beta + c\gamma + \text{etc.}$ is a positive quantity smaller than unity (at least not greater). The same holds for the quantity $a'\alpha' + b'\beta' + c'\gamma' + \text{etc.}$, which is found to be equal to the sum

$$\begin{aligned} &(a'\alpha + b'\beta + c'\gamma + \text{etc.})^2 + (a'\alpha' + b'\beta' + c'\gamma' + \text{etc.})^2 \\ &\quad + (a'\alpha'' + b'\beta'' + c'\gamma'' \text{ etc.})^2 + \text{etc.} \end{aligned}$$

and similarly $a''\alpha'' + b''\beta'' + c''\gamma'' + \text{etc.}$ will be smaller than unity, and so on. Hence $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$ is necessarily smaller than π. *Second*, we have $\Sigma(a\alpha + b\beta + c\gamma + \text{etc.}) = \rho$, since $\Sigma a\alpha = 1, \Sigma b\beta = 1, \Sigma c\gamma = 1$ etc.; from this it is readily deduced that the sum of squares $\Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)$ is greater than $\frac{\rho\rho}{\pi}$, or at least not smaller. Thus the term

$$\frac{\nu^4 - 3\mu^4}{(\pi - \rho)^2} . [\rho - \Sigma((a\alpha + b\beta + c\gamma + \text{etc.})^2)]$$

necessarily lies between the bounds $-\frac{\nu^4 - 3\mu^4}{\pi - \rho}$ and $\frac{\nu^4 - 3\mu^4}{\pi - \rho} . \frac{\rho}{\pi}$ or, if we prefer wider ones, between $-\frac{\nu^4 - 3\mu^4}{\pi - \rho}$ and $+\frac{\nu^4 - 3\mu^4}{\pi - \rho}$, and therefore the square of the mean error to be feared in the value of $\mu\mu = \frac{M}{\pi - \rho}$ lies between the bounds $\frac{2\nu^4 - 4\mu^4}{\pi - \rho}$ and $\frac{2\mu^4}{\pi - \rho}$, so that any degree of precision can be attained, provided only that the number of observations is sufficiently large.

It is very remarkable that under the hypothesis (art. 9, III) on which the theory of least squares was formerly based, that term disappears altogether;

and just as, to obtain an approximate value of the mean error of the observations μ, in all cases the sum $\lambda\lambda + \lambda'\lambda' + \lambda''\lambda'' + \text{etc.} = M$ must be treated as if it were a sum of $\pi - \rho$ errors arising by chance, so under that hypothesis the precision of this determination itself also becomes equal to that which we found should be assigned to a determination from $\pi - \rho$ true errors in art. 15.

SUPPLEMENT

TO THE THEORY

OF THE COMBINATION OF OBSERVATIONS

SUBJECT TO THE SMALLEST ERRORS

BY

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SUPPLEMENT TO THE THEORY OF THE COMBINATION OF OBSERVATIONS SUBJECT TO THE SMALLEST ERRORS.

1.

In the treatment of the theory of the combination of observations included in Volume V of the Commentationes Recentiores, we assumed that the quantities whose values are given by observations lacking absolute precision depend on certain unknown elements in such a way that they are represented as given functions of these elements, and that the main task is to derive these elements from the observations as accurately as possible.

In most cases, indeed, this assumption applies immediately. In other cases, however, the problem presents itself in a slightly different form, so that at first sight it seems doubtful how it can be reduced to the required form. It often happens, namely, that the quantities to which the observations refer have not yet been represented as functions of certain elements, and do not even seem reducible to such a form, at least not conveniently or without circuitous procedures; while, on the other hand, the nature of the matter supplies certain conditions which the true values of the observed quantities must satisfy exactly and necessarily.

Nevertheless, on closer consideration, it is easily seen that this case does not in fact differ essentially from the other, but can be reduced to it. For if the number of observed quantities is denoted by π and the number of conditional equations by σ, and $\pi - \sigma$ of the former are chosen arbitrarily, nothing prevents us from taking these themselves as the elements and considering the others, whose number will be σ, as functions of them by means of the conditional equations; in this way the matter will have been reduced to our assumption.

Yet even if this route leads quite conveniently to the intended goal in very many cases, it cannot be denied that it is less natural, and that it is therefore worthwhile to treat the problem separately in this other form, all the more so because it admits a very elegant solution. Indeed, since this new solution leads to quicker calculations than the solution of the problem in its earlier form whenever σ is smaller than $\frac{1}{2}\pi$, or, what is the same, whenever the number of elements, denoted in the preceding paper by ρ, is greater than $\frac{1}{2}\pi$, it will be appropriate in such a case to prefer the new solution, which we shall explain in the present paper, to the earlier one, provided that the conditional equations can be drawn from the nature of the problem without circuitous procedures.

2.

Let us denote by v, v', v'' etc. the quantities, π in number, whose values become known through observation, and let an unknown quantity depend on them in such a way that it is represented by a given function of them, say u; further, let l, l', l'' etc. be the values of the derivatives

$$\frac{\mathrm{d}u}{\mathrm{d}v}, \quad \frac{\mathrm{d}u}{\mathrm{d}v'}, \quad \frac{\mathrm{d}u}{\mathrm{d}v''} \text{ etc.}$$

corresponding to the true values of the quantities v, v', v'' etc. Just as substituting these true values in the function u gives its true value, so, if for $v.\ v', v''$ etc. values differing from the true ones by the errors e, e', e'' etc. respectively are substituted, an erroneous value of the unknown quantity will be obtained, whose error can be written

$$= le + l'e' + l''e'' + \text{etc.}$$

provided, as we shall always assume, that the errors e, e', e'' etc. are so small that (when the function u is not linear) their squares and products may be neglected. And although the sizes of the errors e, e', e'' etc. remain uncertain, the uncertainty inherent in such a determination of the unknown quantity can nevertheless be estimated in general, namely by the mean error to be feared in such a determination, which by the principles of the preceding paper becomes

$$= \sqrt{(llmm + l'l'm'm' + l''l''m''m'' + \text{etc.})}$$

where m, m', m'' etc. denote the mean errors of the observations, or, if each observation is subject to equal uncertainty,

$$= m\sqrt{(ll + l'l' + l''l'' + \text{etc.})}$$

It is clear that in this calculation it is equally legitimate to take for l, l', l'' etc. the values of the derivatives corresponding to the observed values of the quantities v, v', v'' etc.

3.

Whenever the quantities v, v', v'' etc. are completely independent of one another, the unknown quantity can be determined from them in only one way; consequently, in that case the uncertainty can neither be avoided nor diminished in any way, and nothing is left to arbitrary choice concerning the value of the unknown quantity to be deduced from the observations.

But the situation is very different whenever there is a mutual dependence among the quantities v, v', v'' etc., which we shall assume to be expressed by σ conditional equations

$$X = 0, \quad Y = 0, \quad Z = 0 \text{ etc.}$$

where X, Y, Z etc. denote given functions of the indeterminates v, v', v'' etc. In this case our unknown quantity can be determined in infinitely many different ways by combinations of the quantities v, v', v'' etc., since clearly, in place of the function u, any other function U may be taken such that $U - u$ vanishes identically when $X = 0,\ Y = 0,\ Z = 0$ etc. are set.

In applying this to a particular case, no difference in the value of the unknown quantity would result if the observations had absolute precision; but insofar as they remain subject to errors, a different combination will clearly in general give a different value of the unknown quantity. For example, in place of the error

$$le + l'e' + l''e'' + \text{etc.}$$

which the function u had produced, we shall now have

$$Le + L'e' + L''e'' + \text{etc.}$$

if we take the function U and denote the values of the derivatives $\frac{\mathrm{d}U}{\mathrm{d}v}, \frac{\mathrm{d}U}{\mathrm{d}v'}, \frac{\mathrm{d}U}{\mathrm{d}v''}$ etc. respectively by L, L', L'' etc. And although we cannot specify the errors themselves, it will nevertheless be possible to compare the mean errors to be feared in the different combinations of observations; the best combination will be that in which this mean error becomes as small as possible. Since it becomes

$$= \sqrt{(LLmm + L'L'm'm' + L''L''m''m'' + \text{etc.})}$$

the aim will be to make the sum $LLmm + L'L'm'm' + L''L''m''m'' + \text{etc.}$ attain its smallest value.

4.

Since the infinite variety of functions U which, under the condition stated in the preceding article, can take the place of u need be considered here only insofar as they give rise to different systems of values of the coefficients L, L', L'' etc., it will first of all be necessary to investigate the relation that

must hold among all admissible systems. Let us denote the fixed values of the partial derivatives

$$\begin{array}{llll} \frac{\mathrm{d}X}{\mathrm{d}v}, & \frac{\mathrm{d}X}{\mathrm{d}v'}, & \frac{\mathrm{d}X}{\mathrm{d}v''} & \text{etc.} \\ \frac{\mathrm{d}Y}{\mathrm{d}v}, & \frac{\mathrm{d}Y}{\mathrm{d}v'}, & \frac{\mathrm{d}Y}{\mathrm{d}v''} & \text{etc.} \\ \frac{\mathrm{d}Z}{\mathrm{d}v}, & \frac{\mathrm{d}Z}{\mathrm{d}v'}, & \frac{\mathrm{d}Z}{\mathrm{d}v''} & \text{etc. etc.} \end{array}$$

which they take when the true values are assigned to v, v', v'' etc., respectively by

$$\begin{array}{llll} a, & a', & a'' & \text{etc.} \\ b, & b', & b'' & \text{etc.} \\ c, & c', & c'' & \text{etc. etc.} \end{array}$$

It is clear that if the quantities v, v', v'' etc. are conceived to receive increments $\mathrm{d}v, \mathrm{d}v', \mathrm{d}v''$ etc. which leave X, Y, Z etc. unchanged, so that each remains $= 0$, i.e. increments satisfying the equations

$$\begin{aligned} 0 &= a\mathrm{d}v + a'\mathrm{d}v' + a''\mathrm{d}v'' + \text{etc.} \\ 0 &= b\mathrm{d}v + b'\mathrm{d}v' + b''\mathrm{d}v'' + \text{etc.} \\ 0 &= c\mathrm{d}v + c'\mathrm{d}v' + c''\mathrm{d}v'' + \text{etc. etc.} \end{aligned}$$

then $u - U$ must also remain unchanged, and therefore

$$0 = (l - L)\mathrm{d}v + (l' - L')\mathrm{d}v' + (l'' - L'')\mathrm{d}v'' + \text{etc.}$$

From this it is easily concluded that the coefficients L, L', L'' etc. must be represented by formulas such as

$$\begin{aligned} L &= l + ax + by + cz + \text{etc.} \\ L' &= l' + a'x + b'y + c'z + \text{etc.} \\ L'' &= l'' + a''x + b''y + c''z + \text{etc.} \end{aligned}$$

etc., where x, y, z etc. denote fixed multipliers. Conversely, it is clear that if a system of fixed multipliers x, y, z etc. is taken arbitrarily, a function U can always be specified whose corresponding values of L, L', L'' etc. conform to these equations, and which can take the place of u under the condition stated in the preceding article; indeed, this can be done in infinitely many different ways. The simplest way will be to set $U = u + xX + yY + zZ + \text{etc.}$; more generally, one may set $U = u + xX + yY + zZ + \text{etc.} + u'$, where u' denotes a function of the indeterminates v, v', v'' etc. which always vanishes for $X = 0,\ Y = 0,\ Z = 0$ etc., and whose value in the particular case under consideration becomes a maximum or minimum. But for our purpose this makes no difference.

5.

It will now be easy to assign to the multipliers x, y, z etc. such values that the sum

$$LLmm + L'L'm'm' + L''L''m''m'' + \text{etc.}$$

attains its smallest value. Clearly, for this purpose it is not necessary to know the absolute magnitudes of the mean errors m, m', m'' etc.; their relative proportions suffice. We shall therefore introduce in their place the weights of the observations p, p', p'' etc., that is, numbers inversely proportional to the squares $mm, m'm', m''m''$ etc., taking the weight of an arbitrarily chosen observation as unity. The quantities x, y, z etc. must therefore be determined so that the polynomial in the indeterminates

$$\frac{(ax + by + cz + \text{etc.} + l)^2}{p} + \frac{(a'x + b'y + c'z + \text{etc.} + l')^2}{p'} + \frac{(a''x + b''y + c''z + \text{etc.} + l'')^2}{p''} + \text{etc.}$$

attains its smallest value, which we shall suppose to occur for *determinate* values x^0, y^0, z^0 etc.

Introducing the following notation

$$\frac{aa}{p} + \frac{a'a'}{p'} + \frac{a''a''}{p''} + \text{etc.} = [aa]$$

$$\frac{ab}{p} + \frac{a'b'}{p'} + \frac{a''b''}{p''} + \text{etc.} = [ab]$$

$$\frac{ac}{p} + \frac{a'c'}{p'} + \frac{a''c''}{p''} + \text{etc.} = [ac]$$

$$\frac{bb}{p} + \frac{b'b'}{p'} + \frac{b''b''}{p''} + \text{etc.} = [bb]$$

$$\frac{bc}{p} + \frac{b'c'}{p'} + \frac{b''c''}{p''} + \text{etc.} = [bc]$$

$$\frac{cc}{p} + \frac{c'c'}{p'} + \frac{c''c''}{p''} + \text{etc.} = [cc]$$

etc. and also

$$\frac{al}{p} + \frac{a'l'}{p'} + \frac{a''l''}{p''} + \text{etc.} = [al]$$

$$\frac{bl}{p} + \frac{b'l'}{p'} + \frac{b''l''}{p''} + \text{etc.} = [bl]$$

$$\frac{cl}{p} + \frac{c'l'}{p'} + \frac{c''l''}{p''} + \text{etc.} = [cl]$$

etc.

the condition for a minimum clearly requires that

$$\left.\begin{aligned} 0 &= [aa]x^0 + [ab]y^0 + [ac]z^0 + \text{etc.} + [al] \\ 0 &= [ab]x^0 + [bb]y^0 + [bc]z^0 + \text{etc.} + [bl] \\ 0 &= [ac]x^0 + [bc]y^0 + [cc]z^0 + \text{etc.} + [cl] \\ &\text{etc.} \end{aligned}\right\} \quad (1)$$

After the quantities x^0, y^0, z^0 etc. have been derived from these equations by elimination, we shall set

$$\left.\begin{aligned} ax^0 + by^0 + cz^0 + \text{etc.} + l &= L \\ a'x^0 + b'y^0 + c'z^0 + \text{etc.} + l' &= L' \\ a''x^0 + b''y^0 + c''z^0 + \text{etc.} + l'' &= L'' \\ &\text{etc.} \end{aligned}\right\} \quad (2)$$

With this done, the function of the quantities v, v', v'' etc. that is best suited to determining our unknown quantity and subject to the least uncertainty will be the one whose partial derivatives, in the particular case under consideration, have the values L, L', L'' etc. respectively; and the weight of this determination, which we shall denote by P, will be

$$= \frac{1}{\frac{LL}{p} + \frac{L'L'}{p'} + \frac{L''L''}{p''} + \text{etc.}} \quad (3)$$

or $\frac{1}{P}$ will be the value of the polynomial given above for the system of values of the quantities x, y, z etc. that satisfies equations (1).

6.

In the preceding article we showed how to identify the function U that serves to give the most suitable determination of our unknown quantity; let us now see what *value* the unknown quantity attains in this way. Let this value be denoted by K; it therefore results from substituting in U the observed values of the quantities v, v', v'' etc.; let the same substitution give the function u the value k; finally, let κ be the true value of the unknown quantity, which would therefore result from the true values of the quantities v, v', v'' etc. if we could substitute these in either U or u. Thus we shall have

$$k = \kappa + le + l'e' + l''e'' + \text{etc.}$$
$$K = \kappa + Le + L'e' + L''l'' + \text{etc.}$$

and consequently

$$K = k(L-l)e + (L'-l')e' + (L''-l'')e'' + \text{etc.}$$

Substituting in this equation for $L-l,\ L'-l',\ L''-l''$ etc. the values from (2), and setting

$$\left.\begin{aligned} ae + a'e' + a''e'' + \text{etc.} &= \mathfrak{A} \\ be + b'e' + b''e'' + \text{etc.} &= \mathfrak{B} \\ ce + c'e' + c''e'' + \text{etc.} &= \mathfrak{C} \end{aligned}\right\} \quad (4)$$

etc., we shall have

$$K = k + \mathfrak{A}x^0 + \mathfrak{B}y^0 + \mathfrak{C}z^0 \text{ etc.} \qquad (5)$$

We cannot, indeed, calculate the values of the quantities $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ etc. by formulas (4), since the errors e, e', e'' etc. remain unknown; but it is immediately clear that they are simply the values of the functions X, Y, Z etc. that result when the observed values are substituted for v, v', v'' etc. In this way the system of equations (1), (3), (5) gives the complete solution of our problem, since the remark we made at the end of Art. 2. about basing the calculation of the quantities l, l', l'' etc. on the observed values of the quantities v, v', v'' etc. clearly applies with equal force to the calculation of the quantities a, a', a'' etc. b, b', b'' etc. etc.

7.

In place of formula (3), which expresses the weight of the most plausible determination, several others can be given that will be worth developing.

First we observe that if equations (2) are multiplied respectively by $\frac{a}{p}$, $\frac{a'}{p'}$, $\frac{a''}{p''}$ etc. and added, the result is

$$[aa]x^0 + [ab]y^0 + [ac]z^0 + \text{etc.} = \frac{aL}{p} + \frac{a'L'}{p'} + \frac{a''L''}{p''} + \text{etc.}$$

The left side becomes $= 0$; by analogy, we denote the right side by $[aL]$; we therefore have

$$[aL] = 0, \quad \text{and in exactly the same way} \quad [bL] = 0,\ [cL] = 0 \text{ etc.}$$

Further, multiplying equations (2) in turn by $\frac{L}{p}$, $\frac{L'}{p'}$, $\frac{L''}{p''}$ etc. and adding, we find

$$\frac{lL}{p} + \frac{l'L'}{p'} + \frac{l''L''}{p''} + \text{etc.} = \frac{LL}{p} + \frac{L'L'}{p'} + \frac{L''L''}{p''} + \text{etc.}$$

from which we obtain a *second* expression for the weight,

$$P = \frac{1}{\frac{lL}{p} + \frac{l'L'}{p'} + \frac{l''L''}{p''} + \text{etc.}}$$

Finally, multiplying equations (2) in turn by $\frac{l}{p}$, $\frac{l'}{p'}$, $\frac{l''}{p''}$ etc. and adding, we arrive at a *third* expression for the weight

$$P = \frac{1}{[al]x^0 + [bl]y^0 + [cl]z^0 + \text{etc.} + [ll]}$$

if, in keeping with the other notation, we set

$$\frac{ll}{p} + \frac{l'l'}{p'} + \frac{l''l''}{p''} + \text{etc.} = [ll]$$

From this, with the aid of equations (1), it is easy to pass to a *fourth expression*, which we give as follows:

$$\begin{aligned}\frac{1}{P} = {} & [ll] - [aa]x^0x^0 - [bb]y^0y^0 - [cc]z^0z^0 - \text{etc.} \\ & - 2[ab]x^0y^0 - 2[ac]x^0z^0 - 2[bc]y^0z^0 - \text{etc.}\end{aligned}$$

8.

The general solution that we have developed so far is chiefly suited to the case in which *one* unknown quantity depending on observed quantities is to be determined. But whenever the most plausible values of several unknown quantities depending on the same observations are required, or whenever it is still uncertain which particular unknown quantities ought to be derived from the observations, it will be appropriate to prepare the observations in another way, which we now proceed to develop.

We shall regard the quantities x, y, z etc. as indeterminates, set

$$\left.\begin{aligned} [aa]x + [ab]y + [ac]z + \text{etc.} &= \xi \\ [ab]x + [bb]y + [bc]z + \text{etc.} &= \eta \\ [ac]x + [bc]y + [cc]z + \text{etc.} &= \zeta \end{aligned}\right\} \quad (6)$$

etc., and suppose that elimination gives

$$\left.\begin{aligned} [\alpha\alpha]\xi + [\alpha\beta]\eta + [\alpha\gamma]\zeta + \text{etc.} &= x \\ [\beta\alpha]\xi + [\beta\beta]\eta + [\beta\gamma]\zeta + \text{etc.} &= y \\ [\gamma\alpha]\xi + [\gamma\beta]\eta + [\gamma\gamma]\zeta + \text{etc.} &= z \end{aligned}\right\} \quad (7)$$

etc.

Here it must first of all be observed that coefficients placed symmetrically must be equal, for example

$$\begin{aligned} [\beta\alpha] &= [\alpha\beta] \\ [\gamma\alpha] &= [\alpha\gamma] \\ [\gamma\beta] &= [\beta\gamma] \\ &\text{etc.} \end{aligned}$$

This follows at once from the general theory of elimination in linear equations, but it will also be proved directly below without that theory.

We shall therefore have

$$\left.\begin{aligned} x^0 &= -[\alpha\alpha].[al] - [\alpha\beta].[bl] - [\alpha\gamma].[cl] - \text{etc.} \\ y^0 &= -[\alpha\beta].[al] - [\beta\beta].[bl] - [\beta\gamma].[cl] - \text{etc.} \\ z^0 &= -[\alpha\gamma].[al] - [\beta\gamma].[bl] - [\gamma\gamma].[cl] - \text{etc.} \end{aligned}\right\} \quad (8)$$

$$\text{etc.}$$

from which, if we set

$$\left.\begin{aligned} [\alpha\alpha]\mathfrak{A} + [\alpha\beta]\mathfrak{B} + [\alpha\gamma]\mathfrak{C} + \text{etc.} &= A \\ [\alpha\beta]\mathfrak{A} + [\beta\beta]\mathfrak{B} + [\beta\gamma]\mathfrak{C} + \text{etc.} &= B \\ [\alpha\gamma]\mathfrak{A} + [\beta\gamma]\mathfrak{B} + [\gamma\gamma]\mathfrak{C} + \text{etc.} &= C \end{aligned}\right\} \quad (9)$$

etc., we obtain

$$K = k - A[al] - B[bl] - C[cl] - \text{etc.}$$

or if we further set

$$\left.\begin{aligned} aA + bB + cC + \text{etc.} &= p\varepsilon \\ a'A + b'B + c'C + \text{etc.} &= p'\varepsilon' \\ a''A + b''B + c''C + \text{etc.} &= p''\varepsilon'' \end{aligned}\right\} \quad (10)$$

etc., we shall have

$$K = k - l\varepsilon - l'\varepsilon' - l''\varepsilon'' - \text{etc.} \qquad (11)$$

9.

Comparison of equations (7), (9) shows that the auxiliary quantities A, B, C etc. are the values of the indeterminates x, y, z etc. corresponding to these values of the indeterminates ξ, η, ζ etc.: $\xi = \mathfrak{A},\ \eta = \mathfrak{B},\ \zeta = \mathfrak{C}$ etc. It is therefore clear that we have

$$\left.\begin{aligned} [aa]A + [ab]B + [ac]C + \text{etc.} &= \mathfrak{A} \\ [ab]A + [bb]B + [bc]C + \text{etc.} &= \mathfrak{B} \\ [ac]A + [bc]B + [cc]C + \text{etc.} &= \mathfrak{C} \end{aligned}\right\} \quad (12)$$

etc. Thus, multiplying equations (10) respectively by $\frac{a}{p},\ \frac{a'}{p'},\ \frac{a''}{p''}$ etc.· and adding, we obtain

and in exactly the same way

$$\left.\begin{aligned} \mathfrak{A} &= a\varepsilon + a'\varepsilon' + a''\varepsilon'' + \text{etc.} \\ \mathfrak{B} &= b\varepsilon + b'\varepsilon' + b''\varepsilon'' + \text{etc.} \\ \mathfrak{C} &= c\varepsilon + c'\varepsilon' + c''\varepsilon'' + \text{etc.} \end{aligned}\right\} \quad (13)$$

etc. Now since $\mathfrak{A}$ is the value of the function X when the observed values are substituted for v, v', v'' etc., it will readily be seen that if the corrections $-\varepsilon,\ -\varepsilon',\ -\varepsilon''$ etc. are respectively applied to these, the function X will thereby acquire the value 0, and likewise the functions Y, Z etc. will thereby be reduced to a vanishing value. In a similar way it follows from equation (11) that K is the value of the function u resulting from the same substitution.

We shall call the application of the corrections $-\varepsilon, -\varepsilon', -\varepsilon''$ etc. to the observations the *adjustment of the observations*; we have thus clearly been led to a conclusion of the greatest importance, namely, that the observations, adjusted in the way we have shown, satisfy all the conditional equations exactly, and give to any quantity depending in any way on the observations the very same value that would result from the most suitable combination of the unchanged observations. Since it is therefore impossible to recover the errors themselves e, e', e'' etc. from the conditional equations, whose number is insufficient, we have at least obtained the *most plausible errors*, a name by which we may designate the quantities $\varepsilon, \varepsilon', \varepsilon''$ etc.

10.

Since the number of observations is assumed to be greater than the number of conditional equations, infinitely many other systems satisfying the conditional equations can be found besides the system of most plausible corrections $-\varepsilon, -\varepsilon', -\varepsilon''$ etc.; it is worth investigating how these are related to that sys-

tem. Thus let $-E, -E', -E''$ etc. constitute such a system, different from the most plausible one; we shall have

$$\begin{aligned} aE + a'E' + a''E'' + \text{etc.} &= \mathfrak{A} \\ bE + b'E' + b''E'' + \text{etc.} &= \mathfrak{B} \\ cE + c'E' + c''E'' + \text{etc.} &= \mathfrak{C} \end{aligned}$$

etc. Multiplying these equations respectively by A, B, C etc. and adding, we obtain with the aid of equations (10)

$$p\varepsilon E + p'\varepsilon'E' + p''\varepsilon''E'' + \text{etc.} = A\mathfrak{A} + B\mathfrak{B} + C\mathfrak{C} + \text{etc.}$$

In exactly the same way, however, equations (13) give

$$p\varepsilon\varepsilon + p'\varepsilon'\varepsilon' + p''\varepsilon''\varepsilon'' + \text{etc.} = A\mathfrak{A} + B\mathfrak{B} + C\mathfrak{C} + \text{etc.} \quad (14)$$

Combining these two equations, we easily deduce

$$\begin{aligned} &pEE + p'E'E' + p''E''E'' + \text{etc.} \\ &= p\varepsilon\varepsilon + p'\varepsilon'\varepsilon' + p''\varepsilon''\varepsilon'' + \text{etc.} \\ &\qquad + p(E-\varepsilon)^2 + p'(E'-\varepsilon')^2 + p''(E''-\varepsilon'')^2 + \text{etc.} \end{aligned}$$

The sum $pEE + p'E'E' + p''E''E'' + \text{etc.}$ will therefore necessarily be *greater* than the sum $p\varepsilon\varepsilon + p'\varepsilon'\varepsilon' + p''\varepsilon''\varepsilon'' + \text{etc.}$, which may be stated as a

THEOREM. *The sum of the squares of the corrections by which the observations can be made to agree with the conditional equations, with each square multiplied by the weight of the corresponding observation, is a minimum if the most plausible corrections are adopted.*

This is precisely the principle of least squares, from which equations (12), (10) can also easily be derived directly. Moreover, for this smallest sum, which we shall denote below by S, equation (14) gives us the expression $\mathfrak{A}A + B\mathfrak{B} + C\mathfrak{C} + \text{etc.}$

11.

Since the determination of the most plausible errors is independent of the coefficients l, l', l'' etc., it clearly provides the most convenient preparation for whatever use one may wish to make of the observations. Furthermore, it is clear that this task does not require *indefinite* elimination, or knowledge of the coefficients $[\alpha\alpha], [\alpha\beta]$ etc.; all that is required is to derive the auxiliary quantities A, B, C etc., which we shall hereafter call the *correlates* of the conditional equations $X = 0,\ Y = 0,\ Z = 0$ etc., from equations (12) by definite elimination and substitute them in formulas (10).

Although this method leaves nothing to be desired whenever only the most plausible values of quantities depending on the observations are required, the situation nevertheless seems to be different whenever the weight of a determination is also desired. For this purpose, depending on which of the four expressions given above one wishes to use, knowledge of the quantities L, L', L'' etc., or at least of the quantities x^0, y^0, z^0 etc., seems necessary. It will therefore be useful to investigate elimination more closely, which will open to us an easier way of finding the weights as well.

12.

The relation among the quantities occurring in this investigation is considerably clarified by introducing the function of the second degree in the indeterminates

$$\begin{aligned}&[aa]xx + 2[ab]xy + 2[ac]xz + \text{etc.}\\ &\quad + [bb]yy + 2[bc]yz + \text{etc.} + [cc]zz + \text{etc.}\end{aligned}$$

which we shall denote by T. First, it is immediately evident that this function becomes

$$\frac{(ax + by + cz + \text{etc.})^2}{p} + \frac{(a'x + b'y + c'z + \text{etc.})^2}{p'} + \frac{(a''x + b''y + c''z + \text{etc.})^2}{p''} + \text{etc.} \quad (15)$$

Further, it is clear that

$$T = x\xi + y\eta + z\zeta + \text{etc.} \qquad (16)$$

and if here x, y, z etc. are again expressed in terms of ξ, η, ζ etc. with the aid of equations (7),

$$\begin{aligned}T = {}&[\alpha\alpha]\xi\xi + 2[\alpha\beta]\xi\eta + 2[\alpha\gamma]\xi\zeta + \text{etc.}\\ &+ [\beta\beta]\eta\eta + 2[\beta\gamma]\eta\zeta + \text{etc.}\\ &+ [\gamma\gamma]\zeta\zeta + \text{etc.}\end{aligned}$$

The theory developed above contains two systems of determinate values of the quantities x, y, z etc. and ξ, η, ζ etc.; to the first, in which $x = x^0, y = y^0, z = z^0$ etc. and $\xi = -[al],\ \eta = -[bl],\ \zeta = -[cl]$ etc., there will correspond the following value of T:

$$T = [ll] - \frac{1}{P}$$

This is immediately evident either from the third expression for the weight P, compared with equation (16), or from the fourth; to the second, in which $x = A, y = B, z = C$ etc. and $\xi = \mathfrak{A},\ \eta = \mathfrak{B},\ \zeta = \mathfrak{C}$ etc., there corresponds the value $T = S$, as is evident either from formulas (10) and (15), or from (14) and (16).

13.

The principal task now consists in a transformation of the function T similar to the one we presented in Theoria Motus Corporum Coelestium, Art. 182, and more fully in Disquisitione de elementis ellipticis Palladis. Namely, we shall set (17)

$$[bb,1] = [bb] - \frac{[ab]^2}{[aa]}$$

$$[bc,1] = [bc] - \frac{[ab][ac]}{[aa]}$$

$$[bd,1] = [bd] - \frac{[ab][ad]}{[aa]}$$

etc.

$$[cc,2] = [cc] - \frac{[ac]^2}{[aa]} - \frac{[bc,1]^2}{[bb,1]}$$

$$[cd,2] = [cd] - \frac{[ac][ad]}{[aa]} - \frac{[bc,1][bd,1]}{[bb,1]}$$

etc.

$$[dd,3] = [dd] - \frac{[ad]^2}{[aa]} - \frac{[bd,1]^2}{[bb,1]} - \frac{[cd,2]^2}{[cc,2]}$$

etc. etc. Then, on setting*)

$$[bb,1]y + [bc,1]z + [bd,1]w + \text{etc.} = \eta'$$
$$[cc,2]z. + [cd,2]w + \text{etc.} = \zeta''$$
$$[dd,3]w + \text{etc.} = \varphi'''$$

etc., we shall have

$$T = \frac{\xi\xi}{[aa]} + \frac{\eta'\eta'}{[bb,1]} + \frac{\zeta''\zeta''}{[cc,2]} + \frac{\varphi'''\varphi'''}{[dd,3]} + \text{etc.}$$

*)In the preceding discussion, sets of three letters could suffice for the various systems of quantities referring to the first three conditional equations; here, however, it seemed appropriate to add a fourth, so that the rule of the algorithm might appear more clearly. And since the letters a, b, c; A, B, C; $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ in their natural sequence are naturally followed by $d, D, \mathfrak{D}$, while in the sequence x, y, z the alphabet is exhausted, we have appended w, and likewise to the sequence ξ, η, ζ we have appended φ.

and the quantities $\eta', \zeta'', \varphi'''$ etc. depend on $\xi, \eta, \zeta, \varphi$ etc. through the following equations:

$$\eta' = \eta - \frac{[ab]}{[aa]}\xi$$

$$\zeta'' = \zeta - \frac{[ac]}{[aa]}\xi - \frac{[bc,1]}{[bb,1]}\eta'$$

$$\varphi''' = \varphi - \frac{[ad]}{[aa]}\xi - \frac{[bd,1]}{[bb,1]}\eta' - \frac{[cd,2]}{[cc,2]}\zeta''$$

etc.

All the formulas necessary for our purpose can now easily be obtained from this. Namely, to determine the correlates A, B, C etc., we shall set (18)

$$\mathfrak{B}' = \mathfrak{B} - \frac{[ab]}{[aa]}\mathfrak{A}$$

$$\mathfrak{C}'' = \mathfrak{C} - \frac{[ac]}{[aa]}\mathfrak{A} - \frac{[bc,1]}{[bb,1]}\mathfrak{B}'$$

$$\mathfrak{D}''' = \mathfrak{D} - \frac{[ad]}{[aa]}\mathfrak{A} - \frac{[bd,1]}{[bb,1]}\mathfrak{B}' - \frac{[cd,2]}{[cc,2]}\mathfrak{C}''$$

etc.; then A, B, C, D etc. will be obtained by the following formulas, in reverse order, beginning with the last,

$$\left.\begin{aligned} [aa]A + [ab]B + [ac]C + [ad]D + \text{etc.} &= \mathfrak{A} \\ [bb,1]B + [bc,1]C + [bd,1]D + \text{etc.} &= \mathfrak{B}' \\ [cc,2]C + [cd,2]D + \text{etc.} &= \mathfrak{C}'' \\ [dd,3]D + \text{etc.} &= \mathfrak{D}''' \end{aligned}\right\} \quad (19)$$

etc.

For the sum S, however, we have the new formula (20)

$$S = \frac{\mathfrak{A}\mathfrak{A}}{[aa]} + \frac{\mathfrak{B}'\mathfrak{B}'}{[bb,1]} + \frac{\mathfrak{C}''\mathfrak{C}''}{[cc,2]} + \frac{\mathfrak{D}'''\mathfrak{D}'''}{[dd,3]} \quad \text{etc.}$$

Finally, if the weight P to be assigned to the most plausible determination of the quantity expressed by the function u is desired, we shall set (21)

$$[bl,1] = [bl] - \frac{[ab][al]}{[aa]}$$

$$[cl,2] = [cl] - \frac{[ac][al]}{[aa]} - \frac{[bc,1][bl,1]}{[bb,1]}$$

$$[dl,3] = [dl] - \frac{[ad][al]}{[aa]} - \frac{[bd,1][bl,1]}{[bb,1]} - \frac{[cd,2][cl,2]}{[cc,2]}$$

etc.; this gives (22)

$$\frac{1}{P} = [ll] - \frac{[al]^2}{[aa]} - \frac{[bl,1]^2}{[bb,1]} - \frac{[cl,2]^2}{[cc,2]} - \frac{[dl,3]^2}{[dd,3]} - \text{etc.}$$

Formulas (17). . . .(22), whose simplicity seems to leave nothing to be desired, provide a solution to our problem that is complete in every respect.

14.

After completing the principal problems, we shall also touch on some secondary questions that will shed more light on this subject.

First we must investigate whether the elimination. by which x, y, z etc. must be derived from ξ, η, ζ etc. can ever become impossible. This will clearly happen if the functions ξ, η, ζ etc. are not independent of one another. Let us therefore suppose for the moment that one of them is already determined by the others, so that we have the identity

$$\alpha\xi + \beta\eta + \gamma\zeta + \text{etc.} = 0$$

where α, β, γ etc. denote determinate numbers. We shall therefore have

$$\alpha[aa] + \beta[ab] + \gamma[ac] + \text{etc.} = 0$$
$$\alpha[ab] + \beta[bb] + \gamma[bc] + \text{etc.} = 0$$
$$\alpha[ac] + \beta[bc] + \gamma[cc] + \text{etc.} = 0$$

etc.; hence, if we set

$$\begin{aligned}\alpha a + \beta b + \gamma c + \text{etc.} &= p\theta \\ \alpha a' + \beta b' + \gamma c' + \text{etc.} &= p'\theta' \\ \alpha a'' + \beta b'' + \gamma c'' + \text{etc.} &= p''\theta''\end{aligned}$$

etc., it follows directly that

$$\begin{aligned}a\theta + a'\theta' + a''\theta'' + \text{etc.} &= 0 \\ b\theta + b'\theta' + b''\theta'' + \text{etc.} &= 0 \\ c\theta + c'\theta' + c''\theta'' + \text{etc.} &= 0\end{aligned}$$

etc., and also

$$p\theta\theta + p'\theta'\theta' + p''\theta''\theta'' + \text{etc.} = 0$$

Since all p, p', p'' etc. are by their nature positive quantities, this equation clearly cannot hold unless $\theta = 0,\ \theta' = 0,\ \theta'' = 0$ etc.

Let us now consider the values of the total differentials $\mathrm{d}X, \mathrm{d}Y, \mathrm{d}Z$ etc. corresponding to those values of the quantities v, v', v'' etc. to which the observations relate. By the conclusion we have just reached, these differentials, namely

$$\begin{aligned}a\mathrm{d}v + a'\mathrm{d}v' + a''\mathrm{d}v'' + \text{etc.} \\ b\mathrm{d}v + b'\mathrm{d}v' + b''\mathrm{d}v'' + \text{etc.} \\ c\mathrm{d}v + c'\mathrm{d}v' + c''\mathrm{d}v'' + \text{etc.}\end{aligned}$$

etc., will depend on one another in such a way that, when multiplied respectively by α, β, γ etc., they produce a sum that vanishes identically; or, equivalently, any one of them whose corresponding multiplier α, β, γ etc. is nonzero will itself vanish as soon as all the others are assumed to vanish. Consequently, at least one of the conditional equations $X = 0,\ Y = 0,\ Z = 0$ etc. must be regarded as *redundant*, since it is satisfied automatically as soon as the others are satisfied.

However, if we look more deeply into the matter, it appears that this conclusion by itself holds only for an infinitely small range of variation of the indeterminates. More precisely, two cases must be distinguished: one in which one of the conditional equations $X = 0,\ Y = 0,\ Z = 0$ etc. is already contained in the others absolutely and generally, which can easily be avoided in any case; the other in which, as if by chance, for those concrete values of the quantities v, v', v'' etc. to which the observations relate, one of the functions X, Y, Z etc., for example the first, X, acquires a maximum or minimum value (or, more generally, a stationary value) with respect to all the changes we could make to the quantities v, v', v'' etc. while keeping the equations $Y = 0,\ Z = 0$ etc. satisfied. Yet since in our investigation the

variation of the quantities is considered only within limits so narrow that it can be treated as infinitely small, this second case (which will hardly ever occur in practice) will have the same effect as the first: namely, one of the conditional equations will have to be rejected as redundant. We can be certain that, if all the retained conditional equations are independent of one another in the sense intended here, elimination will necessarily be possible. However, we must reserve a fuller investigation for another occasion; this subject is not undeserving of one on account of its theoretical subtlety rather than its practical usefulness.

15.

In the preceding memoir, Art. 37 and following, we taught a method of determining the precision of observations a posteriori as closely as possible. Namely, if approximate values of π quantities have become known through observations of equal precision, and are compared with the values obtained by calculation from the most plausible values of the ρ elements on which those quantities depend, the squares of the differences must be added and their sum divided by $\pi - \rho$. The resulting quotient may then be regarded as an approximate value of the square of the mean error belonging to this kind of observation. Whenever the observations have unequal precision, these rules need to be changed only in that the squares must be multiplied by the weights of the observations before they are added, and the mean error obtained in this way must refer to observations whose weight has been taken as unity.

Now, in the present treatment, that sum clearly agrees with the sum S, and the difference $\pi - \rho$ with the number of conditional equations σ. Thus, for the mean error of observations whose weight $= 1$, we shall have the expression $\sqrt{\frac{S}{\sigma}}$; this determination will deserve greater confidence the larger the number σ is.

But it will be worthwhile to establish this independently of the preceding investigation as well. For this purpose it will be useful to introduce some new notation. Namely, let the following values of the indeterminates ξ, η, ζ etc.,

$$\xi = a, \quad \eta = b, \quad \zeta = c \quad \text{etc.}$$

correspond to these values of x, y, z etc.,

$$x = \alpha, \quad y = \beta, \quad z = \gamma \quad \text{etc.}$$

so that we have

$$\begin{aligned}
\alpha &= a[\alpha\alpha] + b[\alpha\beta] + c[\alpha\gamma] + \text{etc.} \\
\beta &= a[\alpha\beta] + b[\beta\beta] + c[\beta\gamma] + \text{etc.} \\
\gamma &= a[\alpha\gamma] + b[\beta\gamma] + c[\gamma\gamma] + \text{etc.}
\end{aligned}$$

etc. Likewise, we shall suppose that to the values

$$\xi = a', \quad \eta = b', \quad \zeta = c' \quad \text{etc.}$$

there correspond these,

$$x = \alpha', \quad y = \beta', \quad z = \gamma' \quad \text{etc.}$$

and to these,

$$\xi = a'', \quad \eta = b'', \quad \zeta = c'' \quad \text{etc.}$$

the following,

$$x = \alpha'', \quad y = \beta'', \quad z = \gamma'' \quad \text{etc.}$$

and so on.

With these assumptions, combining equations (4) and (9) gives

$$A = \alpha e + \alpha' e' + \alpha'' e'' + \text{etc.}$$
$$B = \beta e + \beta' e' + \beta'' e'' + \text{etc.}$$
$$C = \gamma e + \gamma' e' + \gamma'' e'' + \text{etc.}$$

etc. Thus, since we have $S = \mathfrak{A}A + \mathfrak{B}B + \mathfrak{C}C + \text{etc.}$, it is clear that

$$\begin{aligned} S = {} & (ae + a'e' + a''e'' + \text{etc.})(\alpha e + \alpha' e' + \alpha'' e'' + \text{etc.}) \\ & + (be + b'e' + b''e'' + \text{etc.})(\beta e + \beta' e' + \beta'' e'' + \text{etc.}) \\ & + (ce + c'e' + c''e'' + \text{etc.})(\gamma e + \gamma' e' + \gamma'' e'' + \text{etc.}) + \text{etc.} \end{aligned}$$

16.

We may regard taking the observations, through which we obtain values of the quantities v, v', v'' etc. affected by random errors e, e', e'' etc., as an experiment which cannot tell us the magnitude of each error committed, but which, by applying the rules explained above, supplies the value of the quantity S, which by the formula just found is a given function of those errors. In such an experiment, some of the random errors may of course turn out larger and others smaller; but the more errors there are, the greater the hope that the value of the quantity S in an individual experiment will deviate little from its mean value. The essential point is therefore to establish the mean value of the quantity S. By the principles presented in the preceding memoir, which it would be superfluous to repeat here, we find this mean value

$$\begin{aligned} = {} & (a\alpha + b\beta + c\gamma + \text{etc.})mm \\ & + (a'\alpha' + b'\beta' + c'\gamma' + \text{etc.})m'm' \\ & + (a''\alpha'' + b''\beta'' + c''\gamma'' + \text{etc.})m''m'' + \text{etc.} \end{aligned}$$

Denoting the mean error of observations whose weight = 1 by μ, so that $\mu\mu = pmm = p'm'm' = p''m''m''$ etc., the expression just found can be written as follows:

$$\left(\frac{a\alpha}{p} + \frac{a'\alpha'}{p'} + \frac{a''\alpha''}{p''} \text{ etc.}\right)\mu\mu + \left(\frac{b\beta}{p} + \frac{b'\beta'}{p'} + \frac{b''\beta''}{p''} + \text{etc.}\right)\mu\mu + \left(\frac{c\gamma}{p} + \frac{c'\gamma'}{p'} + \frac{c''\gamma''}{p''} + \text{etc.}\right)\mu\mu + \text{etc.}$$

But the sum $\frac{a\alpha}{p} + \frac{a'\alpha'}{p'} + \frac{a''\alpha''}{p''} + \text{etc.}$ is found to be

$$= [aa].[\alpha\alpha] + [ab].[\alpha\beta] + [ac].[\alpha\gamma] + \text{etc.}$$

and therefore = 1, as is easily understood from the connection between equations (6) and (7). Likewise,

$$\frac{b\beta}{p} + \frac{b'\beta'}{p'} + \frac{b''\beta''}{p''} + \text{etc.} = 1$$

$$\frac{c\gamma}{p} + \frac{c'\gamma'}{p'} + \frac{c''\gamma''}{p''} + \text{etc.} = 1$$

and so on.

Hence, finally, the mean value of S is $= \sigma\mu\mu$, and thus, insofar as it is permissible to adopt the random value of S as its mean value, we shall have $\mu = \sqrt{\frac{S}{\sigma}}$.

17.

The degree of confidence to be placed in this determination must be judged by the mean error to be feared either in the determination itself or in its square; in the latter case, this mean error is the square root of the mean value of the expression

$$\left(\frac{S}{\sigma} - \mu\mu\right)^2$$

whose expansion follows by reasoning like that set out in the preceding paper, Arts. 39 ff. For the sake of brevity, that reasoning is omitted here, and we give only the formula itself. Namely, the mean error to be feared in the determination of the square $\mu\mu$ is expressed by

$$\sqrt{\left(\frac{2\mu^4}{\sigma} + \frac{\nu^4 - 3\mu^4}{\sigma\sigma} \cdot N\right)}$$

where ν^4 denotes the mean value of the fourth powers of errors. whose weight is $= 1.$, and N the sum

$$\begin{aligned}(a\alpha + b\beta + c\gamma + \text{etc.})^2 + (a'\alpha' + b'\beta' + c'\gamma' + \text{etc.})^2 \\ + (a''\alpha'' + b''\beta'' + c''\gamma'' + \text{etc.})^2\text{etc.}\end{aligned}$$

In general this sum cannot be reduced to a simpler form; but, as can be shown in the same way as in Art. 40 of the preceding paper, its value always lies between π and $\frac{\sigma\sigma}{\pi}$. Under the hypothesis on which the theory of least squares was founded from the outset, the term containing this sum vanishes, since $\nu^4 = 3\mu^4$; and the precision to be attributed to the mean error determined by the formula $\sqrt{\frac{S}{\sigma}}$ is the same as if that mean error had been derived from σ exactly known errors according to Arts. 15 and 16 of the preceding paper.

18.

As we have seen above, two things are required for the adjustment of observations: first, the correlates of the conditional equations, that is, the numbers A, B, C etc. satisfying equations (12), must be derived; second, these numbers must be substituted in equations (10). An adjustment obtained in this way may be called *perfect* or *complete*, to distinguish it from an *imperfect* or *incomplete* adjustment. By the latter term we shall designate adjustments that result from the same equations (10), but with values of the quantities A, B, C etc. substituted that do not satisfy equations (12), that is, values satisfying only some of them or none. As for changes to the observations that cannot be expressed by formulas (10), they are excluded both from the present investigation and from the term "adjustments." Since, insofar as equations (10) apply, equations (13) are wholly equivalent to equations (12), the distinction may also be stated as follows: Completely adjusted observations

satisfy all the conditional equations $X = 0, Y = 0, Z = 0$ etc.; incompletely adjusted observations satisfy either none or at least not all of them. An adjustment, therefore, that satisfies all the conditional equations is necessarily complete.

19.

Since it follows from the very notion of an adjustment that the combination of two adjustments again constitutes an adjustment, it is readily seen that it makes no difference whether the rules by which a perfect adjustment is obtained are applied directly to the original observations or to observations that have already been incompletely adjusted.

Indeed, let $-\theta, -\theta', -\theta''$ etc. be a system of incomplete adjustments obtained from formulas (I):

$$\begin{aligned}\theta p &= A^0 a + B^0 b + C^0 c + \text{etc.}\\ \theta' p' &= A^0 a' + B^0 b' + C^0 c' + \text{etc.}\\ \theta'' p'' &= A^0 a'' + B^0 b'' + C^0 c'' + \text{etc.}\\ &\text{etc.}\end{aligned}$$

Since the observations altered by these adjustments are assumed not to satisfy all the conditional equations, let $\mathfrak{A}^*, \mathfrak{B}^*, \mathfrak{C}^*$ etc. be the values that X, Y, Z etc. take upon their substitution. We must find numbers A^*, B^*, C^* etc. satisfying equations (II):

$$\begin{aligned}\mathfrak{A}^* &= A^*[aa] + B^*[ab] + C^*[ac] + \text{etc.}\\ \mathfrak{B}^* &= A^*[ab] + B^*[bb] + C^*[bc] + \text{etc.}\\ \mathfrak{C}^* &= A^*[ac] + B^*[bc] + C^*[cc] + \text{etc.}\end{aligned}$$

etc. Once this has been done, a complete $\cdot$ adjustment of the observations thus altered is effected by the new changes $-\kappa, -\kappa', -\kappa''$ etc., where $\kappa, \kappa', \kappa''$ etc. are computed from formulas (III):

$$\begin{aligned}\kappa p &= A^* a + B^* b + C^* c + \text{etc.}\\ \kappa' p' &= A^* a' + B^* b' + C^* c' + \text{etc.}\\ \kappa'' p'' &= A^* a'' + B^* b'' + C^* c'' + \text{etc.}\end{aligned}$$

Let us now inquire how these corrections are related to the complete adjustment of the original observations. First, it is evident that

$$\begin{aligned}\mathfrak{A}^* &= \mathfrak{A} - a\theta - a'\theta' - a''\theta'' - \text{etc.}\\ \mathfrak{B}^* &= \mathfrak{B} - b\theta - b'\theta' - b''\theta'' - \text{etc.}\\ \mathfrak{C}^* &= \mathfrak{C} - c\theta - c'\theta' - c''\theta'' - \text{etc.}\end{aligned}$$

etc. Substituting in these equations the values of $\theta, \theta', \theta''$ etc. from (I), as well as the values of $\mathfrak{A}^*, \mathfrak{B}^*, \mathfrak{C}^*$ etc. from (II), we find

$$\mathfrak{A} = (A^0 + A^*)[aa] + (B^0 + B^*)[ab] + (C^0 + C^*)[ac] + \text{etc.}$$
$$\mathfrak{B} = (A^0 + A^*)[ab] + (B^0 + B^*)[bb] + (C^0 + C^*)[bc] + \text{etc.}$$
$$\mathfrak{C} = (A^0 + A^*)[ac] + (B^0 + B^*)[bc] + (C^0 + C^*)[cc] + \text{etc.}$$

etc.; hence it follows that the correlates of the conditional equations satisfying equations (12) are

$$A = A^0 + A^*, \quad B = B^0 + B^*, \quad C = C^0 + C^* \text{ etc.}$$

Equations (10), (I), and (III) therefore show that

$$\varepsilon = \theta + \kappa, \quad \varepsilon' = \theta' + \kappa' \quad \varepsilon'' = \theta'' + \kappa'' \text{ etc.}$$

that is, the complete adjustment of the observations is the same whether computed directly or indirectly, starting from an incomplete adjustment.

20.

Whenever the number of conditional equations is very large, determining the correlates A, B, C etc. by direct elimination may become so lengthy that the person carrying out the calculation may lack the patience for it. It may then often be convenient to obtain a complete adjustment by successive approximations, with the aid of the theorem in the preceding article. Let the conditional equations be divided into two or more classes, and first determine an adjustment by which the equations of the first class are satisfied, neglecting the rest. Next, treat the observations altered by this adjustment so that only the equations of the second class are taken into account. Generally speaking, applying the second system of adjustments will disturb agreement with the equations of the first class; therefore, if only two classes have been made, we shall return to the equations of the first class and determine a third system that satisfies them; then we shall subject the observations, corrected three times, to a fourth adjustment, where only the equations of the second class are considered. Thus, by turns attending now to the first class and now to the second, we shall obtain adjustments that continually decrease; and if the division has been arranged skillfully, after a few iterations we shall reach stable values. If more than two classes have been made, the procedure will be similar: the individual classes will successively enter into the calculation, after the last class, the first will enter the calculation again, and so on. But it suffices here to have indicated this method, whose effectiveness will certainly depend greatly on its skillful application.

21.

It remains for us to complete the proof of the lemma assumed in Art. 8, though for the sake of clarity we shall use other notations better suited to this subject.

Let x^0, x', x'', x''' etc. be indeterminates, and suppose that from the equations

$$\begin{aligned} n^{00}x^0 + n^{01}x' + n^{02}x'' + n^{03}x''' + \text{etc.} &= X^0 \\ n^{10}x^0 + n^{11}x' + n^{12}x'' + n^{13}x''' + \text{etc.} &= X' \\ n^{20}x^0 + n^{21}x' + n^{22}x'' + n^{23}x''' + \text{etc.} &= X'' \\ n^{30}x^0 + n^{31}x' + n^{32}x'' + n^{33}x''' + \text{etc.} &= X''' \\ &\text{etc.} \end{aligned}$$

elimination yields the following:

$$\begin{aligned} N^{00}X^0 + N^{01}X' + N^{02}X'' + N^{03}X''' + \text{etc.} &= x^0 \\ N^{10}X^0 + N^{11}X' + N^{12}X'' + N^{13}X''' + \text{etc.} &= x' \\ N^{20}X^0 + N^{21}X' + N^{22}X'' + N^{23}X''' + \text{etc.} &= x'' \\ N^{30}X^0 + N^{31}X' + N^{32}X'' + N^{33}X''' + \text{etc.} &= x''' \\ &\text{etc.} \end{aligned}$$

Now, substituting in the first and second equations of the second system the values of the quantities X, X', X'', X''' etc. from the first system, we obtain

$$\begin{aligned} x^0 = {} & N^{00}(n^{00}x^0 + n^{01}x' + n^{02}x'' + n^{03}x''' + \text{etc.}) \\ & + N^{01}(n^{10}x^0 + n^{11}x' + n^{12}x'' + n^{13}x''' + \text{etc.}) \\ & + N^{02}(n^{20}x^0 + n^{21}x' + n^{22}x'' + n^{23}x''' + \text{etc.}) \\ & + N^{03}(n^{30}x^0 + n^{31}x' + n^{32}x'' + n^{33}x''' + \text{etc.}) \quad \text{etc.} \end{aligned}$$

and also

$$\begin{aligned} x' = {} & N^{10}(n^{00}x^0 + n^{01}x' + n^{02}x'' + n^{03}x''' + \text{etc.}) \\ & + N^{11}(n^{10}x^0 + n^{11}x' + n^{12}x'' + n^{13}x''' + \text{etc.}) \\ & + N^{12}(n^{20}x^0 + n^{21}x' + n^{22}x'' + n^{23}x''' + \text{etc.}) \\ & + N^{13}(n^{30}x^0 + n^{31}x' + n^{32}x'' + n^{33}x''' + \text{etc.}) \quad \text{etc.} \end{aligned}$$

Since each of these equations must plainly be an identity, in both the former and the latter we may substitute any determinate values whatever for x^0, x', x'', x''' etc. In the former let us substitute

$$x^0 = N^{10}, \quad x' = N^{11}, \quad x'' = N^{12}, \quad x''' = N^{13} \quad \text{etc.}$$

and in the latter

$$x^0 = N^{00}, \quad x' = N^{01}, \quad x'' = N^{02}, \quad x''' = N^{03} \quad \text{etc.}$$

After this, subtraction gives

$$\begin{aligned} N^{10} - N^{01} = {} & (N^{00}N^{11} - N^{10}N^{01})(n^{01} - n^{10}) \\ & + (N^{00}N^{12} - N^{10}N^{02})(n^{02} - n^{20}) \\ & + (N^{00}N^{13} - N^{10}N^{03})(n^{03} - n^{30}) \\ & + \text{etc.} \\ & + (N^{01}N^{12} - N^{11}N^{02})(n^{12} - n^{21}) \\ & + (N^{01}N^{13} - N^{11}N^{03})(n^{13} - n^{31}) \\ & + \text{etc.} \\ & + (N^{02}N^{13} - N^{12}N^{03})(n^{23} - n^{32}) \\ & + \text{etc. etc.} \end{aligned}$$

This equation can also be written as

$$N^{10} - N^{01} = \Sigma(N^{0\alpha}N^{1\beta} - N^{1\alpha}N^{0\beta})(n^{\alpha\beta} - n^{\beta\alpha})$$

where $\alpha\beta$ denote all combinations of unequal indices.

Hence it follows that if

$$\begin{aligned} & n^{01} = n^{10}, \quad n^{02} = n^{20}, \quad n^{03} = n^{30} \\ & n^{12} = n^{21}, \quad n^{13} = n^{31}, \quad n^{23} = n^{32}, \quad \text{etc.} \end{aligned}$$

or, more generally, if $n^{\alpha\beta} = n^{\beta\alpha}$, then we shall also have

$$N^{10} = N^{01}$$

And since the order of the indeterminates in the given equations is arbitrary, under that supposition it is plainly true in general that

$$N^{\alpha\beta} = N^{\beta\alpha}$$

22.

Since the method set out in this memoir finds particularly frequent and convenient application in calculations pertaining to higher geodesy, we hope that readers will welcome an illustration of its rules by some examples drawn from this field.

The conditional equations among the angles of a system of triangles are to be sought chiefly from three sources.

I. The sum of the horizontal angles that complete a full turn of the horizon around the same vertex must equal four right angles.
II. The sum of the three angles in any triangle is equal to a given quantity, since, whenever a triangle lies on a curved surface, the excess of this sum over two right angles can be computed so accurately that it may be treated as absolutely exact.
III. The third source is the ratio of the sides in triangles forming a closed chain. Namely, if a series of triangles is connected so that the second triangle has one side a in common with the first and another, b, with the third; likewise, if the fourth triangle has a side c in common with the third and a side d in common with the fifth, and so on up to the last triangle, whose side in common with the preceding triangle is k, and with the first triangle again is l, then the values of the ratios $\frac{a}{l}, \frac{b}{a}, \frac{c}{b}, \frac{d}{c} \ldots . \frac{l}{k}$ will be known, respectively, from pairs of angles in successive triangles opposite their common sides, by known methods. Since the product of those fractions must be $= 1$, a conditional equation between the sines of those angles will result (with these angles respectively diminished by the third part of the spherical or spheroidal excess, if the triangles lie on a curved surface).

In more complicated systems of triangles, however, it often happens that more conditional equations of the second and third kinds present themselves than may be retained, since some of them are already contained in the others. The rarer case is that equations similar to the conditional equations of the second kind, concerning figures with more sides, must be added, namely only then; when polygons are formed that have not been divided into triangles by measurement. But these matters, which are too remote from the present purpose, we shall discuss more fully on another occasion. Still, we cannot omit a caution: if our theory is to be applied purely and rigorously, it presupposes that the quantities denoted by v, v', v'' etc. are either actually observed directly or derived from observations in such a way that they remain independent of one another, or at least may be regarded as such. In ordinary practice the angles of the triangles themselves are observed, and so may be taken as v, v', v'' etc. But we must remember that if the system also contains triangles whose angles have not been observed directly, but arise as sums or differences of angles actually observed, those angles must not be counted among the observed quantities; their composite form must instead be retained in the calculations. Matters are different in an observing method like that followed by the distinguished STRUVE (Astronomische Nachrichten II, p. 431), in which the directions of the individual sides issuing from the same vertex are obtained by comparison with one and the same arbitrary direction. In that case these very angles are to be taken as v, v', v'' etc.; thus all the angles of the triangles appear in the form of differences, and the conditional equations of the first kind, which are automatically satisfied by the nature of

the case, cease to be needed. The observing method that I myself followed in the triangulation completed in earlier years differs from both the first and the second method, but in its effect it can be equated with the latter, so that at each station the directions of the sides issuing from it, counted from a starting direction chosen, as it were, arbitrarily, must be taken as the quantities v, v', v'' etc. We shall now work through two examples, one belonging to the first method and the other to the second.

23.

The work by the distinguished DE KRAYENHOF, *Précis historique des opérations trigonométriques faites en Hollande*, will furnish our first example. We shall subject to adjustment that part of the network of triangles which lies among the nine points Harlingen, Sneek, Oldeholtpade, Ballum, Leeuwarden, Dockum, Drachten, Oosterwolde, and Gröningen. Between these points nine triangles are formed, designated in that work by the numbers 121, 122, 123, 124, 125, 127, 128, 131, and 132. Their angles, distinguished by the indices specified by us, were observed as follows, according to the table on pp. 77–81:

Triangle 121.

0.	Harlingen	50^0	$58'$	$15''238$
1.	Leeuwarden . . .	82	47	15,351
2.	Ballum	46	14	27,202

Triangle 122.

3.	Harlingen	51	5	39,717
4.	Sneek	70	48	33,445
5.	Leeuwarden . . .	58	5	48,707

Triangle 123.

6.	Sneek	49	30	40,051
7.	Drachten	42	52	59,382
8.	Leeuwarden . . .	87	36	21,057

Triangle 124.

9.	Sneek	45	36	7,492
10.	Oldeholtpade . . .	67	52	0,048
11.	Drachten	66	31	56,513

Triangle 125.

12.	Drachten	53	55	24,745
13.	Oldeholtpade . . .	47	48	52,580
14.	Oosterwolde . . .	78	15	42,347

Triangle 127.

15.	Leeuwarden . . .	59	24	0,645
16.	Dockum	76	34	9,021
17.	Ballum	44	1	51,040

Triangle 128.

18.	Leeuwarden . . .	72	6	32,043
19.	Drachten	46	53	27,163
20.	Dockum	61	0	4,494

Triangle 131.

21.	Dockum	57	1	55,292
22.	Drachten	83	33	14,515
23.	Gröningen	39	24	52,397

Triangle 132.

24.	Oosterwolde . . .	81^0	$54'$	$17''447$
25.	Gröningen	31	52	46,094
26.	Drachten	66	12	57.246

An examination of the connections among these triangles shows that among the 27 angles whose approximate values have become known through observation there are 13 conditional equations: namely, two of the first kind, nine of the second, and two of the third. It is unnecessary to write all these equations here in their finite form, since the calculations require only the quantities denoted in the general theory by $\mathfrak{A}, a, a', a''$, etc., $\mathfrak{B}, b, b', b''$, etc. We therefore write at once, in their place, the equations designated above by (13), which exhibit those quantities; instead of the symbols $\varepsilon, \varepsilon', \varepsilon''$, etc., we shall simply write (0), (1). (2), etc.

In this way, the following equations correspond to the two conditional equations of the first kind:

$$(1) + (5) + (8) + (15) + (18) = -2''197$$
$$(7) + (11) + (12) + (19) + (22) + (26) = -0''436$$

We next find the spheroidal excesses of the nine triangles to be: $1''749$; $1''147$; $1''243$; $1''698$; $0''873$; $1''167$; $1''104$; $2''161$; $1''403$. Thus the first conditional equation of the second kind is the following*): $v^{(0)} + v^{(1)} + v^{(2)} -$

*). In this example we prefer to express the indices with Arabic numerals.

$180^0\,0'\,1''749 = 0$, and the others are obtained in the same way; hence we have the following nine equations:

$$\begin{aligned}
(0) + (1) + (2) &= -3''958 \\
(3) + (4) + (5) &= +0{,}722 \\
(6) + (7) + (8) &= -0{,}753 \\
(9) + (10) + (11) &= +2{,}355 \\
(12) + (13) + (14) &= -1{,}201 \\
(15) + (16) + (17) &= -0{,}461 \\
(18) + (19) + (20) &= +2{,}596 \\
(21) + (22) + (23) &= +0{,}043 \\
(24) + (25) + (26) &= -0{,}616
\end{aligned}$$

The conditional equations of the third kind are more conveniently presented in logarithmic form; the first is therefore

$$\begin{aligned}
&\log\sin(v^{(0)} - 0''583) - \log\sin(v^{(2)} - 0''583) \\
&\quad - \log\sin(v^{(3)} - 0''382) + \log\sin(v^{(4)} - 0''382) \\
&\quad - \log\sin(v^{(6)} - 0''414) + \log\sin(v^{(7)} - 0''414) \\
&\quad - \log\sin(v^{(16)} - 0''389) + \log\sin(v^{(17)} - 0''389) \\
&\quad - \log\sin(v^{(19)} - 0''368) + \log\sin(v^{(20)} - 0''368) = 0
\end{aligned}$$

It seems unnecessary to write the other one in finite form. The following equations correspond to these two equations; each coefficient refers to the seventh figure of the Briggsian logarithms:

$$\begin{aligned}
&17{,}068(0) - 20{,}174(2) - 16{,}993(3) + 7{,}328(4) \\
&\quad - 17{,}976(6) + 22{,}672(7) - 5{,}028(16) + 21{,}780(17) \\
&\quad - 19{,}710(19) + 11{,}671(20) = -371
\end{aligned}$$

$$\begin{aligned}
&17{,}976(6) - 0{,}880(8) - 20{,}617(9) + 8{,}564(10) \\
&\quad - 19{,}082(13) + 4{,}375(14) + 6{,}798(18) - 11{,}671(20) \\
&\quad + 13{,}657(21) - 25{,}620(23) - 2{,}995(24) + 33{,}854(25) = +370
\end{aligned}$$

Since no reason has been indicated for assigning unequal weights to the observations, we shall set $p^{(0)} = p^{(1)} = p^{(2)}$etc. $= 1$. Denoting the correlates of the conditional equations, in the order in which we have presented the cor-

responding equations, by $A, B, C, D, E, F, G, H, I, K, L, M, N$, the following equations result for determining them:

$$
\begin{aligned}
-2''197 &= 5A + C + D + E + H + I + 5{,}917N \\
-0{,}436 &= 6B + E + F + G + I + K + L + 2{,}962M \\
-3{,}958 &= A + 3C - 3{,}106M \\
+0{,}722 &= A + 3D - 9{,}665M \\
-0{,}753 &= A + B + 3E + 4{,}696M + 17{,}096N \\
+2{,}355 &= B + 3F - 12{,}053N \\
-1{,}201 &= B + 3G - 14{,}707N \\
-0{,}461 &= A + 3H + 16{,}752M \\
+2{,}596 &= A + B + 3I - 8{,}039M - 4{,}874N \\
+0{,}043 &= B + 3K - 11{,}963N \\
-0{,}616 &= B + 3L + 30{,}859N \\
-371 &= +2{,}962B - 3{,}106C - 9{,}665D + 4{,}696E \\
&\quad + 16{,}752H - 8{,}039I + 2902{,}27M - 459{,}33N \\
+370 &= +5{,}917A + 17{,}096E - 12{,}053F - 14{,}707G \\
&\quad - 4{,}874I - 11{,}963K + 30{,}859L - 459{,}33M + 3385{,}96N
\end{aligned}
$$

By elimination we obtain:

$A = -0{,}598$	$H = +0{,}659$
$B = -0{,}255$	$I = +1{,}050$
$C = -1{,}234$	$K = +0{,}577$
$D = +0{,}086$	$L = -1{,}351$
$E = -0{,}477$	$M = -0{,}109792$
$F = +1{,}351$	$N = +0{,}119681$
$G = +0{,}271$	

Finally, the most probable errors are given by the formulas

$$
\begin{aligned}
(0) &= C + 17{,}068M \\
(1) &= A + C \\
(2) &= C - 20{,}174M \\
(3) &= D - 16{,}993M
\end{aligned}
$$

etc., whence we obtain the following numerical values; for comparison, we append, with their signs changed, the corrections applied by the distinguished DE KRAYENHOF to the observations:

	DE KR.		DE KR.
(0) = −3″108	−2″090	(14) = +0″795	+2″400
(1) = −1,832	+0,116	(15) = +0,061	+1,273
(2) = +0,981	−1,982	(16) = +1,211	+5,945
(3) = +1,952	+1,722	(17) = −1,732	−7,674
(4) = −0.719	+2,848	(18) = +1,265	+1,876
(5) = −0,512	−3,848	(19) = +2,959	+6,251
(6) = +3,648	−0,137	(20) = −1,628	−5,530
(7) = −3,221	+1,000	(21) = +2,211	+3,486
(8) = −1,180	−1,614	(22) = +0,322	−3,454
(9) = −1,116	0	(23) = −2,489	0
(10) = +2,376	+5.928	(24) = −1,709	+0,400
(11) = +1,096	−3,570	(25) = +2,701	+2.054
(12) = +0,016	+2,414	(26) = −1,606	−3,077
(13) = −2,013	−6,014		

The sum of the squares of our adjustments is found to be = 97,8845. Hence the mean error, insofar as it can be inferred from the 27 observed angles, is

$$= \sqrt{\frac{97{,}8845}{13}} = 2''7440$$

The sum of the squares of the changes that the distinguished DE KRAYENHOF himself applied to the observed angles is found to be = 341,4201.

24.

The triangles among five points of the Hanover triangulation—Falkenberg, Breithorn, Hauselberg, Wulfsode, and Wilsede—will furnish a second example. The directions were observed[*]:

At station *Falkenberg*

0.	Wilsede	187^0	47′	30″311
1.	Wulfsode	225	9	39,676
2.	Hauselberg . . .	266	13	56,239
3.	Breithorn	274	14	43,634

[*] The reference directions to which the individual directions are referred are here regarded as arbitrary, although in fact they coincide with the meridian lines of the stations. The observations will be published in full later; for the present, the figure is found in *Astronomische Nachrichten*, Vol. I, p. 441.

At station *Breithorn*

4.	Falkenberg . . .	94	33	40,755
5.	Hauselberg . . .	122	51	23,054
6.	Wilsede	150	18	35,100

At station *Hauselberg*

7.	Falkenberg . . .	86	29	6,872
8.	Wilsede	154	37	9,624
9.	Wulfsode	189	2	56,376
10.	Breithorn	302	47	37,732

At station *Wulfsode*

11.	Hauselberg . . .	9	5	36,593
12.	Falkenberg . . .	45	27	33,556
13.	Wilsede	118	44	13,159

At station *Wilsede*

14.	Falkenberg . . .	7^0	$51'$	$1''027$
15.	Wulfsode	298	29	49,519
16.	Breithorn	330	3	7,392
17.	Hauselberg . . .	334	25	26,746

These observations allow seven triangles to be formed.

Triangle I.

Falkenberg	8^0	$0'$	$47''395$
Breithorn	28	17	42,299
Hauselberg	143	41	29,140

Triangle II.

Falkenberg	86	27	13,323
Breithorn	55	44	54,345
Wilsede	37	47	53.635

Triangle III.

Falkenberg	41	4	16,563
Hauselberg	102	33	49,504
Wulfsode	36	21	56,963

Triangle IV.

Falkenberg	78	26	25,928
Hauselberg	68	8	2,752
Wilsede	35	25	34,281

Triangle V.

Falkenberg	37	22	9,365
Wulfsode	73	16	39,603
Wilsede	69	21	11,508

Triangle VI.

Breithorn	27	27	12,046
Hauselberg	148	10	28,108
Wilsede	4	22	19,354

Triangle VII.

Hauselberg	34^0	$25'$	$46''752$
Wulfsode	109	38	36,566
Wilsede	35	55	37,227

There will therefore be seven conditional equations of the second kind (equations of the first kind are evidently absent). To derive them, we must first compute the spheroidal excesses of the seven triangles. For this purpose, the absolute magnitude of at least one side must be known: the side between Wilsede and Wulfsode is 22377,94 metres. The spheroidal excesses of triangles I, II, III, IV, V, VI, and VII are therefore, in order: I. . . $0''202$; II. . . $2''442$; III. . . $1''257$; IV. . . $1''919$; V. . . $1''957$; VI. . . $0''321$; VII. . . $1''295$.

Now, if the directions are denoted by $v^{(0)}, v^{(1)}, v^{(2)}, v^{(3)}$ etc., in the order given above and distinguished by the indices, the angles of triangle I are

$$v^{(3)} - v^{(2)}, \quad v^{(5)} - v^{(4)}, \quad 360^0 + v^{(7)} - v^{(10)}$$

and hence the first conditional equation is

$$-v^{(2)} + v^{(3)} - v^{(4)} + v^{(5)} + v^{(7)} - v^{(10)} + 179^0\,59'\,59''798 = 0$$

The other six triangles likewise furnish further equations. A little attention, however, shows that these seven equations are not independent: the second is identical with the sum of the first, fourth, and sixth, and the sum of the third and fifth is identical with the sum of the fourth and seventh. We shall therefore omit the second and fifth. In place of the remaining conditional equations in finite form, we write the corresponding equations from system (13), using (0), (1), (2), etc. for the symbols $\varepsilon, \varepsilon'$, etc.:

$$\begin{aligned}
-1''368 &= -(2) + (3) - (4) + (5) + (7) - (10) \\
+1{,}773 &= -(1) + (2) - (7) + (9) - (11) + (12) \\
+1{,}042 &= -(0) + (2) - (7) + (8) + (14) - (17) \\
-0{,}813 &= -(5) + (6) - (8) + (10) - (16) + (17) \\
-0{,}750 &= -(8) + (9) - (11) + (13) - (15) + (17)
\end{aligned}$$

Eight conditional equations of the third kind could be obtained from the system of triangles, since triples could be formed both from the four triangles I, II, IV, and VI and from the following III, IV, V, and VII for this purpose. A little attention nevertheless shows that *two* suffice, one from each group, since the others must already be contained in these and the preceding conditional equations. The sixth conditional equation will therefore be

$$\begin{aligned}&\log\sin(v^{(3)} - v^{(2)} - 0''067) - \log\sin(v^{(5)} - v^{(4)} - 0''067)\\ &+ \log\sin(v^{(14)} - v^{(17)} - 0''640) - \log\sin(v^{(2)} - v^{(0)} - 0''640)\\ &+ \log\sin(v^{(6)} - v^{(5)} - 0''107) - \log\sin(v^{(17)} - v^{(16)} - 0''107) = 0\end{aligned}$$

and the seventh

$$\begin{aligned}&\log\sin(v^{(2)} - v^{(1)} - 0''419) - \log\sin(v^{(12)} - v^{(11)} - 0''419)\\ &+ \log\sin(v^{(14)} - v^{(17)} - 0''640) - \log\sin(v^{(2)} - v^{(0)} - 0''640)\\ &+ \log\sin(v^{(13)} - v^{(11)} - 0''432) - \log\sin(v^{(17)} - v^{(15)} - 0''432) = 0\end{aligned}$$

with the corresponding equations from system (13) being

$$\begin{aligned}+25 = &+ 4{,}31(0) - 153{,}88(2) + 149{,}57(3) + 39{,}11(4) - 79{,}64(5)\\ &+ 40{,}53(6) + 31{,}90(14) + 275{,}39(16) - 307{,}29(17)\\ -3 = &+ 4{,}31(0) - 24{,}16(1) + 19{,}85(2) + 36{,}11(11) - 28{,}59(12)\\ &- 7{,}52(13) + 31{,}90(14) + 29{,}06(15) - 60{,}96(17)\end{aligned}$$

If we now assign the same degree of certainty to each direction, setting $p^{(0)} = p^{(1)} = p^{(2)}$ etc. $= 1$, and denote by A, B, C, D, E, F, G the correlates of the seven conditional equations in the order followed here, their determination is to be obtained from the following equations:

$$\begin{aligned}-1{,}368 = &+ 6A - 2B - 2C - 2D + 184{,}72F - 19{,}85G\\ +1{,}773 = &- 2A + 6B + 2C + 2E - 153{,}88F - 20{,}69G\\ +1{,}042 = &- 2A + 2B + 6C - 2D - 2E + 181{,}00F + 108{,}40G\\ -0{,}813 = &- 2A - 2C + 6D + 2E - 462{,}51F - 60{,}96G\\ -0{,}750 = &+ 2B - 2C + 2D + 6E - 307{,}29F - 133{,}65G\\ +25 = &+ 184{,}72A - 153{,}88B + 181{,}00C - 462{,}51D - 307{,}29E\\ &+ 224868F + 16694{,}1G\\ -3 = &- 19{,}85A - 20{,}69B + 108{,}40C - 60{,}96D - 133{,}65E\\ &+ 16694{,}1F + 8752{,}39G\end{aligned}$$

By elimination we deduce

$$\begin{aligned} A &= -0{,}225 \\ B &= +0{,}344 \\ C &= -0{,}088 \\ D &= -0{,}171 \\ E &= -0{,}323 \\ F &= +0{,}000215915 \\ G &= -0{,}00547462 \end{aligned}$$

The most probable errors are now given by the formulas:

$$\begin{aligned} (0) &= -C + 4{,}31F + 4{,}31G \\ (1) &= -B - 24{,}16G \\ (2) &= -A + B + C - 153{,}88F + 19{,}85G \end{aligned}$$

etc., from which the numerical values follow

$(0) = +0''065$	$(9) = +0''021$
$(1) = -0{,}212$	$(10) = +0{,}054$
$(2) = +0{,}339$	$(11) = -0{,}219$
$(3) = -0{,}193$	$(12) = +0{,}501$
$(4) = +0{,}233$	$(13) = -0{,}282$
$(5) = -0{,}071$	$(14) = -0{,}256$
$(6) = -0{,}162$	$(15) = +0{,}164$
$(7) = -0{,}481$	$(16) = +0{,}230$
$(8) = +0{,}406$	$(17) = -0{,}139$

The sum of the squares of these errors is found to be = 1,2288; hence the mean error of one direction, insofar as it can be inferred from the 18 observed directions, is

$$= \sqrt{\frac{1{,}2288}{7}} = 0''4190$$

25.

To illustrate the other part of our theory as well by an example, we investigate the precision with which the side Falkenberg–Breithorn is determined from the side Wilsede–Wulfsode with the aid of adjusted observations. The function u by which the former is expressed in this case is

$$u = 22877^{\mathrm{m}}94 \times \frac{\sin(v^{(13)} - v^{(12)} - 0''652).\sin(v^{(14)} - v^{(16)} - 0''814)}{\sin(v^{(1)} - v^{(0)} - 0''652).\sin(v^{(6)} - v^{(4)} - 0''814)}$$

Its value, obtained from the corrected values of the directions $v^{(0)}, v^{(1)}$ etc., is

$$= 26766^{\mathrm{m}}68$$

Differentiating that expression gives the following, if the differentials $\mathrm{d}v^{(0)}$, $\mathrm{d}v^{(1)}$ etc. are taken to be expressed in seconds of arc:

$$\begin{aligned}\mathrm{d}u = {} & 0^{\mathrm{m}}16991(\mathrm{d}v^{(0)} - \mathrm{d}v^{(1)}) + 0^{\mathrm{m}}08836(\mathrm{d}v^{(4)} - \mathrm{d}v^{(6)}) \\ & - 0^{\mathrm{m}}03899(\mathrm{d}v^{(12)} - \mathrm{d}v^{(13)}) + 0^{\mathrm{m}}16731(\mathrm{d}v^{(14)} - \mathrm{d}v^{(16)})\end{aligned}$$

From this we further find

$$\begin{aligned}[al] &= -0{,}08836 \\ [bl] &= +0{,}13092 \\ [cl] &= -0{,}00260 \\ [dl] &= +0{,}07895 \\ [el] &= +0{,}03899 \\ [fl] &= -40{,}1315 \\ [gl] &= +10{,}9957 \\ [ll] &= +0{,}13238\end{aligned}$$

Finally, by the methods given above, taking the metre as the unit of linear dimensions, we find

$$\frac{1}{P} = 0{,}08329, \quad \text{or} \quad P = 12{,}006$$

whence the mean error to be expected in the value of the side Falkenberg–Breithorn is $= 0{,}2886m$ metres (where m is the mean error to be expected in the observed directions, expressed in seconds of arc); accordingly, if we adopt the value of m derived above,

$$= 0^{\mathrm{m}}1209$$

Moreover, inspection of the system of triangles readily shows that the point Hauselberg could have been omitted from it altogether, while the connection between the sides Wilsede–Wulfsode and Falkenberg–Breithorn remained intact. But it is contrary to sound method to *discard* observations on that account when they relate to the point Hauselberg[*)], since they can certainly contribute to increased precision. To make clearer how much increase in precision results from this, we repeated the calculation, excluding everything

[*)] Most of these observations had already been made before the point Breithorn was discovered and incorporated into the system.

that relates to the point Hauselberg; in this way eight of the 18 directions given above are removed, and the most probable errors of the remainder are found as follows:

(0) = +0″327	(12) = +0″206
(1) = −0,206	(13) = −0,206
(3) = −0,121	(14) = +0,327
(4) = +0,121	(15) = +0,206
(6) = −0,121	(16) = +0,121

The value of the side Falkenberg–Breithorn then comes out to be = 26766^{m}63, differing little from the value derived above; the calculation of the weight, however, gives

$$\frac{1}{P} = 0{,}13082 \quad \text{or} \quad P = 7{,}644$$

and hence the mean error to be expected is = 0,36169m metres = 0^{m}1515. It is therefore evident that, through the addition of observations; which relate to the point Hauselberg, the weight of the determination of the side Falkenberg–Breithorn has increased in the ratio of 7,644 to 12,006, or of unity to 1.571.

NOTICES.

Göttingische gelehrte Anzeigen. 26 February 1821.

On 15 February, a lecture was submitted to the Royal Society by Councillor Gauss, entitled

Theoria Combinationis observationum erroribus minimis obnoxiae, pars prior,

and dealing with one of the most important applications of the calculus of probability. No observations concerning the determination of magnitudes in the natural world, however accurately they may be made and however excellent the instruments used, can ever have *absolute* accuracy: they always remain approximations, subject to greater or smaller errors. The errors considered here are not those whose sources are exactly known and whose magnitude can each time be calculated for particular observations; such errors can and should be subtracted from the observed quantities, which is equivalent to their not being present at all. The situation is quite different for errors to be regarded as random, arising from the limited sharpness of the senses, from various unavoidable and irregular imperfections of instruments, and from external disturbances that act irregularly (at least as far as we can tell)—for example, the movement of the atmosphere when looking, or the lack of absolute stability when instruments are set up. These random errors, which cannot be subjected to calculation, cannot be *removed*; by careful attention and by multiplying the observations, the observer can only *reduce* them. Once the observer has done what is possible, however, it is the geometer's task to assess, by strict mathematical principles, the uncertainty of the observations and of the quantities derived from them by calculation. Most important, wherever quantities connected with the observations can be derived from them by different combinations, the geometer must prescribe the method that leaves the least uncertainty to be feared.

Although random errors as such follow no law, and occur without order—larger in one observation, smaller in another—it is nevertheless certain that, for a specified kind of observation and with the observer and instruments also regarded as fixed, the errors arising from each simple source of error are not merely confined within certain limits: every possible error between those limits also has its own relative probability, so that errors of different sizes are expected more or less often according to their magnitude. Anyone with an exact and complete understanding of the nature of such a source of error would be able to determine these limits and the relation between

the probability of each error and its magnitude, in much the same way that, once the rules of a game of chance are known, one can calculate the limits of possible gains and losses and their relative probabilities. The same holds for the total error arising from the combined action of the simple sources of error. These ideas are not confined to direct observations; they also apply to indirect determinations of magnitudes derived from observations. In practice, of course, we will almost always lack the means to specify *a priori* the law governing the probabilities of errors.

How we are to assess, in general, the inadmissibility of a particular kind of observation depends in part on our discretion. We may take only the magnitude of the greatest possible errors as our measure, or also take account of the greater or lesser probability of each possible error. The latter seems more appropriate. This consideration, however, can be brought to bear in many ways on the matter. Calculators have hitherto used the so-called probable (not *most probable*) error as their measure: it is the error beyond which all possible errors together have just as much probability as all those on this side of it. For this purpose, however, it will be *far more advantageous* to use the *mean* error instead of the probable error, provided that this still-indeterminate notion is defined in the right way. Assign each error a moment depending on its magnitude; multiply the moment of each possible error by its probability and add the products. The error whose moment equals this aggregate must then be regarded as the mean error. Which function of the magnitude of the error we choose for its moment remains, in turn, a matter of our *discretion*, provided that the function is always positive and is greater for larger errors than for smaller ones. The author chose the simplest such function, namely the square; this choice also has several other highly important advantages that no other choice has. Otherwise, any other power with an even exponent could be used; the larger the exponent, the closer one would come to the principle in which only the greatest errors serve as the measure of accuracy. Against the way in which a great geometer defined the mean error—equating the moments of errors to the errors themselves when positive, and using their opposites when negative—it may be observed that this conflicts with mathematical continuity, is no less arbitrary than any other choice, and gives results that are far less simple and satisfactory. It also seems more natural in itself for the moment of errors to grow more strongly than the errors themselves, since one would surely rather tolerate the simple error twice than the double error once.

These explanations had to be given first if anything at all was to be said here about the contents of the investigation, of which the present treatise forms the first part.

When the quantities whose values are found by observation are related in a known way to an equal number of unknown quantities, the values of the

unknown quantities can, generally speaking, be derived from the observations by calculation. Those values, of course, will be only approximately correct, insofar as the observations were; the calculus of probability has nothing to do here except assess the uncertainty of those determinations, taking the uncertainty of the observations as given. If there are more unknown quantities than observations, the unknowns cannot be determined from the observations at all. But if there are fewer unknown quantities than observations, the problem is overdetermined: infinitely many combinations are then possible for deriving the unknown quantities from the observations. These combinations would all have to yield the same results if the observations were absolutely accurate, but under the circumstances they yield results that differ more or less from one another. Choosing, from this infinitely varied set of combinations, the most suitable one—that is, the one for which the uncertainty of the results is as small as possible—is unquestionably one of the most important problems in applying mathematics to the natural sciences.

The author of the present treatise, who first investigated this problem in 1797 on the principles of the calculus of probability, soon found that it was impossible to determine the *most probable* values of the unknown quantities unless the function representing the probability of errors was known. If it is not known, nothing remains but to assume such a function hypothetically. It seemed most natural to him first to take the reverse course and seek the function that must be assumed if it is to yield, in the simplest of all cases, a rule generally recognized as good: namely, that the arithmetic mean of several values found by observations of equal reliability for one and the same unknown quantity must be regarded as the most probable. It followed that the probability of an error x must be assumed proportional to an exponential quantity of the form e^{-hhxx}, and that the very method to which he had already been led a few years earlier by other considerations then became generally necessary. He later had occasion to use this method almost daily, especially from 1801 onward, in all kinds of astronomical calculations; LEGENDRE had meanwhile also arrived at it. This method is now in general use under the name method of least squares; its justification by the calculus of probability, as well as the determination of the accuracy of the results themselves and other related investigations, is developed in detail in the *Theoria Motus Corporum Coelestium.*

The Marquis DELAPLACE, who later considered this subject from a new point of view—not by seeking the most probable values of the unknown quantities, but by seeking the most suitable combination of the observations—found the remarkable result that, when the number of observations is regarded as infinitely large, the method of least squares is always the most suitable combination, independently of the function expressing the probability of errors.

It can be seen from this that both grounds for the method still leave something to be desired. The first depends entirely on the hypothetical form of the probability of errors; if that form is rejected, the values of the unknown quantities found by the method of least squares are no longer the most probable, just as the arithmetic means in the simplest case mentioned above are no longer so. The second ground leaves us completely in the dark about what is to be done with a moderate number of observations. In that case, the method of least squares no longer has the standing of a law prescribed by the calculus of probability; it recommends itself only through the simplicity of the operations involved.

In the present treatise, the author has taken up this investigation anew, starting from a point of view similar to DELAPLACE's but defining the mean error to be feared in a different way, which seems to him more natural in itself. He hopes that friends of mathematics will be pleased to see that, on the new foundation given here, the method of least squares appears in general as the most suitable combination of observations—not approximately, but with mathematical exactness—whatever function may describe the probability of errors and whether the number of observations be large or small.

Many other remarkable investigations are closely connected with the main subject, but their scope compelled the author to reserve the development of most of them for a future second lecture. Of those already included in this first part, allow us to state only one result here. If the function expressing the relative probability of each individual error is unknown, then naturally it is also impossible to determine the probability that an error falls between given limits. Nevertheless, provided that larger errors always have a smaller (at least, not a greater) probability than smaller ones, the probability that the error falls between the limits $-x$ and $+x$ must necessarily be greater (at least, not smaller) than $\frac{x}{m}\sqrt{\frac{1}{3}}$, when x is less than $m\sqrt{\frac{3}{4}}$, and not less than $1 - \frac{4mm}{9xx}$, when x is greater than $m\sqrt{\frac{3}{4}}$, where m denotes the mean error to be feared in the observations. For $x = m\sqrt{\frac{3}{4}}$, the two expressions, as one sees, coincide.